

Cecilia Siliansky de Andreazzi

04/24/1984. Rio de Janeiro, RJ, Brazil

Curriculum Vitae

Academic address Fundação Oswaldo Cruz, Presidência da Fiocruz, Campus Fiocruz Mata Atlântica.
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Profile

Associate researcher at Fundação Oswaldo Cruz with interest in the ecology and evolution of species interaction networks. Background in ecology, disease ecology, evolutionary biology, complex systems, mathematical modeling and numerical simulations.

Education

2012 - 2016 Ph.D in Ecology.
Universidade de São Paulo, USP, São Paulo, Brasil
Title: Coevolution in antagonistic species interaction networks: structure and dynamics
Advisor: Paulo Roberto Guimarães Júnior
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)

2015 - 2016 Visiting Ph.D student (6 months)
Swiss Federal Institute of Aquatic Science and Technology
Advisor: Carlos Melián
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)

2006 - 2008 M.Sc in Ecology.
Universidade Federal do Rio de Janeiro, UFRJ, Rio de Janeiro, Brasil
Title: Reproductive phenology and seed dispersal and predation of *Attalea humilis* palm in Atlantic Forest fragments
Advisors: Fernando Antonio dos Santos Fernandez and Alexandra Pires
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)

2002 - 2006 B.Sc in Biological Sciences.
Universidade Federal do Rio de Janeiro, UFRJ, Rio de Janeiro, Brasil
Title: Small mammals population ecology in the Pantanal
Advisor: Paulo Sérgio D'Andrea
Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)

Appointments

2006 - 2016 Research assistant, Fundação Oswaldo Cruz (FIOCRUZ, Rio de Janeiro, Brazil)
2016 - Associate researcher, Fundação Oswaldo Cruz (FIOCRUZ, Rio de Janeiro, Brazil)

Teaching

2017 - *Novel methods for the study of Biodiversity*, at the Instituto Oswaldo Cruz graduate programs (annually offered), Brazil
2017 - *Population Ecology*, at the Instituto Oswaldo Cruz graduate programs (annually offered), Brazil

Juries

Examiner at 1 PhD jury, 2 PhD qualifications and 2 Bachelor juries.

Complementary Education

2014 - 2014	Complex Systems Summer School Santa Fé Institute, New Mexico, United States
2013 - 2013	II Southern-Summer School on Mathematical Biology. ICTP South American Institute for Fundamental Research, ICTP-SAIFR, Brazil
2013 - 2013	Energetic Approach to Food Webs. ICTP South American Institute for Fundamental Research, ICTP-SAIFR, Brazil
2012 - 2012	Food Webs Dynamics. Laboratório Nacional de Computação Científica, LNCC, Petrópolis, Brazil
2007 - 2007	III Ibero-American Course on Frugivory and Seed Dispersal Universidade Estadual Paulista, UNESP RIO CLARO, Brazil

Peer Review

- International Journal of Biodiversity
- Cadernos de Saúde Pública
- Chiroptera Neotropical
- Proceedings of the Royal Society – B
- Oecologia
- Oikos
- Ecography
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Grants & Awards

2016	<i>Best poster</i> , Conference on Mathematical Modeling and Control of Communicable Diseases
2012 -	CNPq - Ph.D. Scholarship
2006 - 2007	CNPq - M.Sc. Scholarship
2006	Research Grant, IdeaWild
2005	<i>The ten best Scientific Initiation works in FIOCRUZ</i> , Fundação Oswaldo Cruz.
2003 - 2006	CNPq - Undergraduate Research Scholarship

Publications

1. MELIÁN, C. J., MATTHEWS, B., **ANDREAZZI, C. S.**, RODRIGUEZ, J. P., HARMON, L. J., FORTUNA, M. A. 2018. Deciphering the Interdependence Between Ecological and Evolutionary Networks. *Trends in Ecology & Evolution*. In press.
2. **ANDREAZZI, C. S.**, GUIMARÃES Jr., P.R., MELIÁN, C. J. 2018. Eco-evolutionary feedbacks promote fluctuating selection and long-term stability of antagonistic networks. *Proceedings of the Royal Society B-Biological Sciences*: 285: 20172596.
3. STELLA, M., SELAKOVIC, S., ANTONIONI, A., **ANDREAZZI, C. S.** 2018. Ecological multiplex interactions determine the role of species for parasite spread amplification. *eLife* 7:e3.

4. RAIMUNDO, R. L. G., MARQUITTI, F. M. D., **ANDREAZZI, C. S.**, PIRES, M. M., GUIMARÃES Jr., P. R. 2017. Ecology and evolution of species-rich interaction networks. In Dáttilo, W. and Rico-Gray, V. (Eds.). Book: Ecological Networks in the Tropics. Springer Publisher.
5. **ANDREAZZI, C. S.**, THOMPSON, J., GUIMARÃES Jr. P. R. 2017. Network structure and selection asymmetry drive coevolution in species-rich antagonistic interactions. *The American Naturalist* 190: 99-115.
6. STELLA, M., **ANDREAZZI, C. S.**, SELAKOVIC, S., ALIREZA, G., ANTONIONI, A. 2017. Parasite Spreading in Spatial Ecological Multiplex Networks. *Journal of Complex Networks* 5(3): 485-511.
7. **ANDREAZZI, C. S.**, PIRES, A. S., PIMENTA, C. S., FERNANDEZ, F. A. S. 2012. Increased female reproduction favours the large-seeded palm *Attalea humilis* in small Atlantic Forest fragments. *Journal of Tropical Ecology* 28: 321-325.
8. **ANDREAZZI, C. S.**, PIMENTA, C. S., PIRES, A. S., FERNANDEZ, F. A. S., OLIVEIRA-SANTOS, L. G. & MENEZES, J. F. S. 2012. Increased Productivity and Reduced Seed Predation Favor a Large-seeded Palm in Small Atlantic Forest Fragments. *Biotropica*, 44: 237-245.
9. **ANDREAZZI, C. S.**, RADEMAKER, V., GENTILE, R., HERRERA, H., JANSEN, A. M., DANDREA, P. S. 2011. Population ecology of small rodents and marsupials in a semi-deciduous tropical forest on the Southeast Pantanal wetland, Brazil. *Zoologia* 28: 762-770.
10. MORATELLI, R., **ANDREAZZI, C. S.**, OLIVEIRA, J. A. & CORDEIRO, J. L. P. 2011. Current and potential distribution of *Myotis simus* (Chiroptera, Vespertilionidae). *Mammalia*, 75: 227-234.
11. **ANDREAZZI, C. S.**, PIRES A. S. & FERNANDEZ, F. A. S. 2009. Mamíferos e palmeiras neotropicais: interações em paisagens fragmentadas. *Oecologia Brasiliensis* 13: 554-574.

Scientific Communication

1. **ANDREAZZI, C. S.**, ASTEGIANO, J., GUIMARÃES Jr. P. R. Mechanisms driving the evolution of species-rich interaction networks In: **International Conference of the Institute for Biocomputation & Physics of Complex Systems**, BIFI, Zaragoza. 2018.
2. **ANDREAZZI, C. S.**, ASTEGIANO, J., GUIMARÃES Jr. P. R. Coevolution by trait matching and exploitation barriers shape the structure of antagonistic and mutualistic networks in different ways In: **Ecological Society of America meeting**, Portland. 2017.
3. **ANDREAZZI, C. S.**, ASTEGIANO, J., GUIMARÃES Jr. P. R. Evolução e Co-evolução em redes antagonistas E mutualistas In: **VI Reunión Binacional de Ecología**, Puerto Iguazú. 2016.
4. **ANDREAZZI, C. S.**, GUIMARÃES Jr. P. R., MELIÁN, C. J. Eco-evolutionary feedbacks promote interaction fluctuating selection and long-term stability of antagonistic networks. In: **Evolution meeting**, Austin. 2016.
5. STELLA, M., **ANDREAZZI, C. S.**, SELAKOVIC, S., ALIREZA, G., ANTONIONI, A. Parasite Spreading in Spatial Ecological Multiplex Networks. In: **Conference on Mathematical Modeling and Control of Communicable Diseases**, Rio de Janeiro. 2016.
6. **ANDREAZZI, C. S.**, GUIMARÃES, P. R., Jr. Generalist enemies reduce the escalation of attack and defense traits in species-rich networks. In: **Ecological networks: theory, empiricism and practice**

in a changing world, Bristol. 2015.

7. **ANDREAZZI, C. S.**, GUIMARÃES, P. R., Jr. Network structure drive coevolutionary dynamics in species-rich antagonistic interactions. In: **Evolution meeting**, Guarujá. 2015.
8. **ANDREAZZI, C. S.**, GUIMARÃES, P. R., Jr. Coevolution in antagonistic networks. In: **Winter Workshop on Complex Systems**, Brussels. 2015.
9. **ANDREAZZI, C. S.**, CORDEIRO, J. L. P., ABREU, A. V. G., FRAGOSO, J. M. V., Oliveira, L. F. B. Experimental evaluation of *Attalea phalerata* seed predation along a gradient of palm density in the Brazilian Pantanal In: **49th Annual Meeting of the Association for Tropical Biology and Conservation**, Bonito, 2012, Bonito.
10. **ANDREAZZI, C. S.**, CORDEIRO, J. L. P., MORATELLI, R. Distribution models for three phylogenetically close related South American species of *Myotis* Kaup (Chiroptera, Vespertilionidae) In: **10th International Mammalogical Congress**, Mendoza, 2009,
11. **ANDREAZZI, C. S.**, PIMENTA, C. S., PIRES, A. S., FERNANDEZ, F. A. S. Dispersão de frutos da palmeira pindoba (*Attalea humilis*) por mamíferos In: **IV Congresso Brasileiro de Mastozoologia**, São Lourenço, 2008.
12. PIRES, A. S., **ANDREAZZI, C. S.**, PIMENTA, C. S., FERNANDEZ, F. A. S., GALETTI, M. Seed dispersal and predation in two Atlantic Forest Palms with different responses to habitat loss In: **Annual Meeting of the Association for Tropical Biology and Conservation: Linking Tropical Biology with Human Dimensions**, Morelia, 2007.
13. **ANDREAZZI, C. S.**, PIMENTA, C. S., PIRES, A. S., FERNANDEZ, F. A. S. Predação de sementes de *Attalea humilis* Mart. Ex. Spreng (Arecaceae) em fragmentos de Mata Atlântica (RJ) de diferentes tamanhos In: **VIII Congresso de Ecologia do Brasil**, Caxambú, 2007.
14. **ANDREAZZI, C. S.**, RADEMAKER, V., JANSEN, A. M., DANDREA, P. S. Ecologia de Populações de pequenos mamíferos numa área preservada do pantanal da Nhicolândia In: **III Congresso Brasileiro de Mastozoologia**, Aracruz, 2005.

CURRICULUM VITÆ: FRANÇOIS MASSOL

Unit Evo-Eco-Paléo, SPICI group
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Born September 17th, 1979
French nationality
Single, three children
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Research group webpage: <http://spici.weebly.com/>

[RESEARCHGATE](#); [MENDELEY](#); [ACADEMIA](#); [ORCID](#); [RESEARCHER ID](#); [GOOGLE SCHOLAR](#); [SCOPUS](#)

Recent professional positions

- 2012 – **Permanent researcher** CNRS, [CEFE](#) Montpellier (2012 – 2014), [EEP](#) Lille (2013 –)
Hired through a national competition, with ~6 posts for ~200 junior applicants in ecology/evolution.
- 2009 – 2010 **Marie Curie post-doc** at the University of Texas, Austin, USA (supervisor: M. Leibold)
- 2008 – 2012 **Permanent researcher** IRSTEA, Aix-en-Provence (2008 – 2009 and 2010 – 2012)
- 2002 – 2008 **Engineer / Civil servant** ENGREF, Paris

Education

- 2015 **Habilitation (HDR)** in Biodiversity Science at the University of Lille
HDR is the diploma allowing researchers to be PhD student main supervisors; ~ tenure or privatdozent.
- 2004 – 2008 **PhD** in Population Biology & Ecology at the University of Montpellier
Supervised by D. Gerdeaux and P. Jarne; paid as civil servant for the Ministry of Agriculture.
- 2003 – 2004 **Master** in Evolutionary Biology & Ecology at the University of Montpellier
- 1999 – 2003 **Interdisciplinary graduate studies** at Ecole Polytechnique and ENGREF, 2 MSc degrees

Advising & supervision

- 2013 – Member of 8 PhD advisory committees (Montpellier, Toulouse, La Réunion, Marseille, Dijon)
- 2008 – Adviser or co-adviser of 7 post-docs, 3 PhD students, 14 MSc students
1 PhD student defended his thesis (F. Laroche) and is now a permanent researcher in France.
5 former post-docs now have permanent or tenure-track positions in Europe, Argentina or the US (K. Allhoff, J. Astegiano, J. Pantel, M. Pantasso, M. Thomas).

Teaching

- 2014 **Invited graduate course** – evolutionary dynamics, SAIIR, São Paulo, Brazil
- 2013 **Invited graduate course** – metacommunity theory, ETH/Uni. Zürich, Ascona, Switzerland
- 2013 – **Regular lectures at MSc level** (10 – 16h per year) – ecological modelling, evolutionary biology, methods for the study of evolution, the publication process, etc., Univ. Lille

Funding

As principal investigator (PI)

- 2009 – 2011 **DEFTER** Post-doctoral fellowship (EU Marie Curie programme) on “dispersal effects on food web functioning: theory and experimental research based on freshwater plankton systems” (in M. Leibold’s laboratory in Austin, TX)
- 2010 – 2012 **FoodWebs** Working group at the National Institute for Mathematical and Biological Synthesis (NIMBioS) on “food web dynamics and stoichiometric constraints in metaecosystems” (PIs: C. Klausmeier, M. Leibold, F. Massol & R. Sterner)
- 2013 – 2018 **MIRES** Working group funded by the Réseau National Systèmes Complexes (RNSC) and the Institut National de la Recherche Agronomique (INRA) applied mathematics and informatics department on “interdisciplinary methods for the study of seed exchange networks” (PIs: P. Barbillon, S. Caillon, F. Massol, M. Thomas & N. Verzealen)
- 2013 – 2014 **DyBRES** Working group funded by the RNSC on “biodiversity dynamics in spatial ecological networks” (PIs: S. Manel, F. Massol & F. Munoz)
- 2014 – 2017 **COREIDS** Working group at the Centre de Synthèse et d’Analyse de la Biodiversité (CESAB) on “Predicting community resilience to invasions from diversity and network structure” (PIs: P. David & F. Massol)
- 2014 **RESEDA** Bilateral cooperation (funded by the University of Lille 1) with the University of Exeter on “selection regimes, dispersal and local adaptation” (PIs: F. Débarre & F. Massol)

2014 – 2018	ARSENIC	Project funded by the Agence Nationale de la Recherche (ANR) on “adaptation and resilience of spatial ecological networks to human-induced changes” (PIs: N. Loeuille & F. Massol)
2016 – 2017	FEEDING	Interregional project Ghent-Louvain-Lille on “functional and evolutionary ecology of dispersal: integrative approaches under global change” (PIs: N. Schtickzelle, D. Bonte & F. Massol)
2017 – 2018	E²-SPACE	Interregional project Ghent-Louvain-Lille on “eco-evolutionary dynamics in space” (PIs: N. Schtickzelle, D. Bonte & F. Massol)

Others

2011 – 2014	NETSEED	Working group at the CESAB on “seed exchange networks” (PI: D. McKey)
2012 – 2017	AFFAIRS	Project funded by the ANR BIOADAPT programme on “adaptation of fragmented freshwater assemblages in rapidly-changing environments” (PI: P. David)
2015 – 2018	AREOLAIRE	Project funded by the Région Nord Pas-de-Calais / FRB programme on “adaptation, regression and expansion at species range margins” (PI: A. Duputié)
2015 – 2016	MADRES	Project funded by the CNRS interdisciplinary mission MoMIS programme on “modelling and analysis of the dynamics of seed exchange networks” (PI: S. Martin)
2018 – 2019	sEcoEvo	Working group at the sDiv synthesis centre (Berlin, Germany) on “Biodiversity Dynamics – The Nexus Between Space & Time” (PIs: R. G. Gillespie & M. Hickerson)
2018 – 2021	SUBANTECO	Project funded by the Institut polaire français Paul-Emile Victor (IPEV) on “Subantarctic biodiversity, effects of climate change and biological invasions on terrestrial biota” (PI: D. Renault)
2017 – 2021	NGB	Project funded by the ANR on “next-generation biomonitoring of change in ecosystem structure and function” (PI: D. Bohan)

Awards

2016 – 2020	Supervision and Research Bonus (PEDR), granted by the CNRS
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Commissions of trust

2017 –	Founder, manager and recommender for <i>Peer Community in Ecology</i>
2017 –	Participation in researcher and lecturer hiring committees (Univ. Paris Diderot, IRSTEA)
2016 – 2017	Guest volume editor for the 56 th and 57 th volumes of <i>Advances in Ecological Research</i>
2015 –	Member of the administrative board of the French Society of Ecology and Evolution (SFE ²)
2015 –	Referee at 1 HDR defence; examiner at 2 HDR defences (Montpellier, Toulouse, Lille)
2014 –	Referee at 1 PhD defence; examiner at 5 PhD defences (Paris, Dijon, Montpellier, Ecole Polytechnique)
2014 –	Subject editor for the journal <i>Oikos</i>
2011 –	Reviewer of proposals for funding agencies (Austria, Belgium, France, Israel, Russia, Switzerland)
2008 –	Reviewer for > 20 journals including <i>PNAS</i> , <i>Proc. R Soc B</i> , <i>Ecol. Lett.</i> , <i>Nat. Commun.</i> , <i>Nat. Ecol. Evol.</i>

Since 2007, I have authored **55 peer-reviewed articles (14 as first author, 9 as last author)**, in major multidisciplinary (1 *Nature*, 2 *Nature Communications*, 3 *Scientific Reports*, 1 *Science Advances*) and general biology, ecology and evolutionary biology journals (2 *Trends in Ecology and Evolution*, 4 *Ecology Letters*, 1 *Biological Reviews*, 1 *Ecological Monographs*, 1 *Methods in Ecology and Evolution*, 2 *Proceedings of the Royal Society B*, 4 *Evolution*, 1 *Journal of Ecology*...). I have also contributed to **1 book chapter** and edited **2 books** (volumes 56 and 57 of the *Advances in Ecological Research* series). I gave **2 invited plenary lectures**, **7 invited oral communications**, **5 oral contributions**, 21 oral presentations in workshops and 16 seminars in various universities.

Citations: 1682 (Google Scholar) / 1104 (ISI); **1 highly cited paper** (on ISI web of science¹)

H-index: 22 (Google Scholar) / 18 (ISI)

All articles can be found here: <https://sites.google.com/a/polytechnique.org/francoismassol/publications>

In the following, students are underlined. * indicates co-first authorship. Impact factors (IF) were obtained from the 2016 ISI Journal Citation Reports, numbers of citations were obtained from Web of Science on 25/05/2018.

¹ received enough citations to place them in the top 1% of their academic field based on a highly cited threshold for the field and publication year.

Representative peer-reviewed publications (non-exhaustive list)

- (1) Gravel, D.*, **Massol, F.***, & Leibold, M. (2016) Stability and complexity in model meta-ecosystems. *Nature Communications*, **7**, 12457. **11 citations; journal IF: 12.124**
- (2) **Massol, F.** & Débarre, F. (2015) Evolution of dispersal in temporally variable environments: the importance of life cycles. *Evolution*, **69**, 1925-1937. **5 citations; journal IF: 4.201**
- (3) **Massol, F.**, Dubart, M., Calcagno, V., Cazelles, K., Jacquet, C., Kéfi, S., & Gravel, D. (2017) Island biogeography of food webs. *Advances in Ecological Research*, **56**, 183-262. **7 citations; journal IF: 5.056**
- (4) **Massol, F.**, Gravel, D., Mouquet, N., Cadotte, M. W., Fukami, T., & Leibold, M. A. (2011) Linking ecosystem and community dynamics through spatial ecology. *Ecology Letters*, **14**, 313-323. **99 citations; journal IF: 9.449**
- (5) Gravel, D., **Massol, F.**, Canard, E., Mouillot, D., & Mouquet, N. (2011) Trophic theory of island biogeography. *Ecology Letters*, **14**, 1010-1016. **77 citations; journal IF: 9.449**
- (6) **Massol, F.**, Duputié, A., David, P., & Jarne, P. (2011) Asymmetric patch size distribution leads to disruptive selection on dispersal. *Evolution*, **65**, 490-500. **32 citations; journal IF: 4.201**
- (7) Laroche, F., Jarne, P., Lamy, T., David, P., & **Massol, F.** (2015) A neutral theory for interpreting correlations between species and genetic diversity in communities. *American Naturalist*, **185**, 59-69. **14 citations; journal IF: 4.167**
- (8) Tasiemski, A., **Massol, F.**, Cuvillier-Hot, V., Boidin-Wichlacz, C., Roger, E., Rodet, F., Fournier, I., Thomas, F., & Salzet, M. (2015) Reciprocal immune benefits based on the complementary production of antibiotics by the leech *Hirudo verbana* and its gut symbiont *Aeromonas veronii*. *Scientific Reports*, **5**, 17498. **7 citations; journal IF: 4.259**
- (9) Cheptou, P.-O. & **Massol, F.** (2009) Pollination fluctuations drive evolutionary syndromes linking dispersal and mating system. *American Naturalist*, **174**, 46-55. **54 citations; journal IF: 4.167**
- (10) Duputié, A. & **Massol, F.** (2013) An empiricist's guide to theoretical predictions on the evolution of dispersal. *Interface Focus*, **3**. **31 citations; journal IF: 2.693**
- (11) Calcagno, V.*, **Massol, F.***, Mouquet, N., Jarne, P., & David, P. (2011) Constraints on food chain length arising from regional metacommunity dynamics. *Proceedings of the Royal Society Biological Sciences Series B*, **278**, 3042-3049. **25 citations; IF: 4.940**
- (12) Schleuter, D., Daufresne, M., **Massol, F.**, & Argillier, C. (2010) A user's guide to functional diversity indices. *Ecological Monographs*, **80**, 469-484. **236 citations; IF: 8.759**

Other representative publications (non-exhaustive lists)

Open review publications

Lavoir, A.-V., Staudt, M., Schnitzler, J. P., Landais, D., **Massol, F.**, Rocheteau, A., Rodriguez-Cortina, R., Zimmer, I., & Rambal, S. (2009) Drought reduced monoterpene emissions from *Quercus ilex* trees: results from a throughfall displacement experiment within a forest ecosystem. *Biogeosciences Discussions*, **6**, 863-893.

Edited books

Bohan, D. A., Dumbrell, A., & **Massol, F.** (2017) *Networks of Invasion: A Synthesis of Concepts*. Vol. 56, *Advances in Ecological Research* series.

Bohan, D. A., Dumbrell, A., & **Massol, F.** (2017) *Networks of Invasion: Empirical Evidence and Case Studies*. Vol. 57, *Advances in Ecological Research* series.

Book chapters

Massol, F., Jabot, F., Manel, S., & Munoz, F. (2015) Etude des réseaux en écologie. In: *Les Réseaux dans les Sociétés Animales* (ed.: Sueur, C.) Editions Matériologiques, 49-98.

Science popularisation & expertise

Massol, F. & Chave, J. (2017) Ecologie fondamentale : les grands défis. In: *Prospectives de l'Institut Ecologie & Environnement du CNRS, Compte-rendu des journées des 22, 23 et 24 Février 2017, Bordeaux*, 7-13.

Invited presentations

Plenary talks at international conferences

- 2016 Networks of interactions in ecology: Everything you always wanted to know about ecological networks (but were afraid to ask). *Société Française d'Ecologie internationale conference*, Marseille, France
- 2014 Individual dispersal, fluxes of nutrients and the stability of ecosystems. *SysBio conference*, Lyon, France

Invited talks at international symposia

- 2017 The evolution of conditional dispersal under different life cycles. *CIMI conference: Ecology and evolutionary biology, deterministic and stochastic models*, Toulouse, France
- 2016 Life-history traits and the properties of metaecosystems: a theoretical perspective. *Per Brink thematic symposium, Nordic Oikos conference*, Turku, Finland
- 2016 Diversification of dispersal rates in variable environments. *Maths-Bio conference day*, Paris, France
- 2015 Joint evolution of dispersal and the allocation of dispersal cost in heterogeneous landscapes. *Mathematical Models in Ecology and Evolution*, Paris, France
- 2014 Community dynamics and the evolution of dispersal. *ECMTB meeting*, Göteborg, Sweden
- 2014 Rescuing metacommunity ecology using random matrix theory. *Journées Modélisation Aléatoire et Statistique*, Toulouse, France
- 2010 Recasting spatial food web ecology as an ecosystem science. *ESA meeting*, Pittsburgh, USA

Other representative presentations (non-exhaustive lists)

Contributed oral communications

- 2017 Eco-evolutionary dynamics of colonization rate, patch occupancy and food chain length in model metacommunities. *ESEB meeting*, Groningen, the Netherlands, with V. Calcagno
- 2017 Effects of cue reliability, environmental uncertainty and the life cycle on the evolution of emigration strategies. *British Ecological Society meeting*, Ghent, Belgium, with F. Débarre
- 2011 Evolution in temporally variable environments: the importance of life cycles. *Mathematical & Theoretical Ecology conference*, Colchester, UK, with F. Débarre

Talks at workshops

- 2017 Modelling the dynamics of spatially structured ecosystems. *Ecosystem Theory conference*, Montpellier
- 2017 Dispersal and the stability of ecosystems. *Hausdorff School: Large random graphs: geometry and applications*, Bonn, Germany
- 2017 Evolution of dispersal in spatially and temporally variable environments: life cycles and informed dispersal. *FEEDING workshop*, Ghent, Belgium
- 2012 Metapopulation-based approaches to spatial food webs. *Networks and interactions in agriculture workshop*, Dijon

Seminars

- 2018 The ecology and evolution of food-chain length in metacommunities. *Sherbrooke, Canada*
- 2017 Evolution of dispersal in spatially and temporally variable environments. *Louvain-la-Neuve, Belgium*
- 2016 Evolution of dispersal in spatially and temporally variable environments: life cycles and informed dispersal. *Helsinki, Finland*
- 2015 Stability, dispersal and ecological networks. *Ecole Polytechnique, Palaiseau, France*
- 2014 A model for the evolution of dispersal in heterogeneous environments. *Ghent, Belgium*
- 2013 Adaptive dynamics in a heterogeneous and uncertain world. *Montpellier, France*
- 2012 The evolution of dispersal: old questions, new answers. *Paris, France*

Julia ASTEGIANO

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Associated researcher of the Ecology of Interspecific Interactions and Conservation Biology Group, Multidisciplinary Institute of Plant Research (IMBIV), National University of Cordoba - CONICET. CC495, Córdoba, Argentina. CP 5000

RECENT TIMELINE

- 1998 – 2003** Biologist. National University of Cordoba, Argentine.
- 2005 – 2010** Ph.D. in Biological Sciences. National University of Cordoba, Argentine.
- 2011 – 2015** Postdoctoral position, Department of Ecology, Institute of Biosciences, University of São Paulo, Brazil.
- 2012 – 2013** Postdoctoral position, Dynamic and adaptation of plant populations, Centre d'Ecologie Fonctionnelle et Evolutive (CEFE), France.
- 2015-2016** Postdoctoral position at the Multidisciplinary Institute of Plant Research.
- 2016-** Associated researcher of CONICET at the Multidisciplinary Institute of Plant Research.

ADVISING AND SUPERVISION

- 2017- Co-adviser of a PhD. Student at the National University of Córdoba, Argentina. Founding: CONICET.

TEACHING

Argentinean School of Mathematical Biology at the National University of Córdoba, Argentina (edition 2016, 2018; professor).

Ecological Networks, at the University of São Paulo, Brazil (editions 2012 and 2014; assistant professor).

Ecology of the Atlantic Forest, at the University of São Paulo, Brazil (edition 2012; professor).

Inquiring Ecology, at the University of São Paulo, Brazil (editions 2012 and 2014; assistant professor).

FUNDING

- 2011-2015** Post-doctoral fellowship (FAPESP) on "robustness of plant-pollinator networks and fragmented landscapes: the role of breeding system". PI: Paulo Guimarães and Julia Astegiano.
- 2012-2013** Post-doctoral fellowship (FAPESP) on "robustness of plant-pollinator networks and fragmented landscapes: the role of breeding system and dispersal strategy on plant species persistence". PI: Paulo Guimarães, Pierre Olivier Cheptou, François Massol, Julia Astegiano.
- 2017-2019** Project founded by FONCyT on "landscape heterogeneity and resilience of pollination services: basis for the ecological intensification of horticultural crops in Argentina". PI: Julia Astegiano.
- 2017-2019** Project founded by FONCyT on "ecosystem services provisioned by *Apis mellifera* and native bees and land use". PI: Ramiro Aguilar.
- 2017-2020** Project founded by CYTED on "the decline of pollinators, pollination

- and plant dispersal ecosystem services in protected areas (SEPODI). PI (Argentina): Lorena Ashworth.
- 2018** Grant from the Argentinean Minister of Science and Technology (CELF program) to organize the VIII School on mathematical biology: “Ecosystems dynamics and mathematical models: contributing to the sustainability of agroecological systems”. PI: Julia Astegiano, Paula Nieto, Lucas Barberis, Silvina Fenoglio, Gustavo Sibona.

PEER REVIEW IN JOURNALS AND SCIENTIFIC REPORTS

Oikos; Evolutionary Ecology; Agriculture, Ecosystems and Environment; Advances in Ecological Research; Scientific reports; Ecography; Revista Colombiana de Entomología; Brazilian Journal of Biology.

Evaluator of the IPBES report on Pollination Services.

Manager and recommender of *Peer Community in Ecology* and recommender of *Peer Community in Evolutionary Ecology*.

JURIES

Examiner at 2 PhD and 2 master juries.

REVIEWER OF PROPOSALS FOR PUBLIC FUNDING AGENCIES

CONICET, UdeLaR (National University of Uruguay).

SCIENTIFIC ANIMATION

Co-organizer of conferences (Argentinean Meeting of Ecology, Córdoba 2006; *Argentinean School of Mathematical Biology, Córdoba 2016, 2018*)

Co-organizer of symposia (Bi-national Meeting of Ecology, Argentina-Chile, Puerto Iguazú 2016)

Member of the administrative board of the Argentinean Ecological Society (AsAE) from 2007 to 2008.

PUBLICATIONS (TOP 5 RELATED TO THE PROPOSED RESEARCH)

Astegiano J, Altermatt F, Massol F (2017). Disentangling the co-structure of multilayer interaction networks: degree distribution and module composition in two-layer bipartite networks. *Scientific reports* 7: 15465.

Astegiano J, Guimaraes Jr P, Cheptou P-O, Morais Vidal M, Mandai, CY, Ashworth, L, Massol F (2015). Persistence of Plants and Pollinators in the face of Habitat Loss: Insights from Trait-based Metacommunity Models. *Advances in Ecological Research* 53:201-248.

Astegiano J, Massol F, Morais Vidal M, Cheptou P-O, Guimaraes Jr PR (2015). The Robustness of Plant-pollinator Assemblages: Linking Plant Interaction Patterns and Sensitivity to Pollinator Loss. *Plos One* 10(2): e0117243.

Bohan DA, Raybould A, Mulder C, Woodward G, Tamaddon-Nezhad S, Bluthgen N, Pocock M.J.O., Muggleton S, Evans DM, **Astegiano J**, Massol F, Loeuille N, Petit S, Macfadyen S (2013). Chapter One - Networking Agroecology: Integrating the Diversity of Agroecosystem Interactions. *Advances in Ecological Research* 49: 1-67.

Astegiano J, Funes G, Galetto L (2013). Plant recruitment success and range size:

transition probabilities from ovule to reproductive individuals in two annual sympatric *Ipomoea*. *Acta Oecologica* 48: 76-82.

PUBLICATION SUMMARY

10 articles in peer-reviewed journals; H-index: 6, 119 citations (Google scholar May 29th, 2018). Six articles as senior author, three invited communications at international symposia and workshops and 28 contributions at national and international conferences.

INVITED PRESENTATIONS

- 2012** **Astegiano J**, Guimarães Jr PR. Plant-pollinator networks and fragmented landscapes: does plants' breeding system matter? 49th Annual Meeting of the Association for Tropical Biology and Conservation (ATBC), Bonito, Brazil. Symposia.
- 2015** **Astegiano J**. Robustness of plant-pollinator networks and plant syndromes: from real communities to metacommunity models. Atelier Réseaux Trophiques, Agrinaples, Paris, France. Workshop.
- 2017** **Astegiano J**. Predicting the vulnerability of the pollination benefit in agroecological ecosystems: networks, functional traits and the metacommunity perspective. 1st Meeting of the SEPDI project, Morelia, México. Symposia.

CURRICULUM VITAE (MAY 2018)

DR. RER. NAT. LUTZ BECKS

ADDRESS: Schillerstrasse 14
24116 Kiel
Phone: +49 (0)151 58533983

INSTITUTIONAL ADDRESS: Max-Planck Institute for Evolutionary Biology
Community Dynamics Group
August-Thienemann-Str. 2, 24306 Plön Germany
Phone: +49 (0)4522 763230, Email: lbecks@evolbio.mpg.de
www.evolbio.mpg.de/comdyn

DATE OF BIRTH: 21. April 1978 PLACE OF BIRTH: Jülich
CITIZENSHIP: Germany

PROFESSIONAL EXPERIENCE

since 10. 2011 Leader of an independent junior research group (*Emmy Noether program and Heisenberg Program, DFG*) at the Max Planck Institute for Evolutionary Biology.

11. 2010 – 10.2011 Postdoctoral Fellow at University of Cologne (Germany) with Prof. Hartmut Arndt (*Volkswagen Foundation Scholarship*).

10.2009 – 10.2010 Postdoctoral Fellow at University of Toronto (Canada) with Prof. Aneil F. Agrawal (*Volkswagen Foundation Scholarship*).

08.2006 – 09.2008 Postdoctoral Fellow at Cornell University, Ithaca NY (USA) with Prof. Nelson G. Hairston Jr. and Prof. Stephen P. Ellner.

EDUCATION

03.2004 – 10.2006 Ph.D. in Zoology at the Zoological Institute, University of Cologne (Germany) with Prof. Hartmut Arndt. Thesis title: "Complex population dynamics in microbial systems" (*magna cum laude*).

02.2002 – 02.2004 Studies and graduation at the University of Cologne (Germany) with a Diploma in Biology. Thesis title: "Chaotic dynamics – a common phenomenon in a two-prey-one-predator system? - Theoretical and experimental analyses of bacteria – protozoan interactions in chemostats" (*sehr gut*).

08.2001 – 01.2002 Erasmus exchange student at the University of Lund (Sweden).

10.1998 – 07.2001 Studies at the University of Cologne (Germany) in Biology.

PUBLICATIONS

Frickel, J., P. Feulner, E. Karakoc & **L. Becks**. *Population size changes and selection drive patterns of parallel evolution in a host-virus system*. **Nature Communications**, 1706.

Huang, W., Traulsen, A., Werner, B., Hiltunen, T. & **L. Becks**. 2017. *Dynamical trade-offs arise from species coevolution and lower the species diversity*. **Nature Communications**, 2059.

Ehrlich, E., **L. Becks** & U. Gaedke, 2017. *The equalizing trade-off: A key to explain species coexistence and co-occurrence*. **Ecology**, 98, 3188-3198.

- Hiltunen, T., Kaitala, V., Laakso, J. & **L. Becks**, 2017. *Evolutionary contribution to coexistence of competitors in microbial food webs*. **Proceedings of the Royal Society B**, 284, 1864.
- Frickel, J., L. Theodosiou & **L. Becks**, 2017. *Rapid evolution of hosts begets species diversity at the cost of intraspecific diversity*. **PNAS**, 201701845.
- Cairns, J., Frickel, J., Jalasvuori, M., Hiltunen, T. & **L. Becks**, 2017. *Genomic evolution of bacterial populations under co-selection by antibiotics and phage*. **Molecular Ecology**, 26, 1848-1859.
- Koch, H. & **L. Becks**, 2017. *The cost of facultative sex in a prey adapting to predation*. **Journal of Evolutionary Biology**, 30, 210-220. (Recommended by Faculty 1000).
- Cairns, J., **Becks, L.**, Jalasvuori, M. & T. Hiltunen, 2016. *Sublethal streptomycin concentrations and lytic bacteriophage interactively promote resistance evolution*. **Phil. Trans. R. Soc. B.**, 372, 20160040.
- Frickel, J., Sieber, M. & **L. Becks**, 2016. *Eco-evolutionary dynamics in a coevolving host-virus system*. **Ecology Letters**, 19, 450-459.
- Haafke, J., Abou Chakra, M. & **L. Becks**, 2016. *Eco-evolutionary dynamics select indirectly for sex in predator populations*. **Evolution**, 70, 641-652.
- Koch, H., Jeschke, A. & **L. Becks**, 2015. *Use of ddPCR in experimental evolution studies*. **Methods in Ecology and Evolution**, 7, 340-351.
- Hiltunen, T. Ayan, G. & **L. Becks**, 2015. *Environmental fluctuations restrict eco-evolutionary dynamics in predator-prey system*. **Proceedings of the Royal Society B**, 282. doi:10.1098/rspb.2015.0013.
- Becks, L.** & Y. Salehi-Alawi, *Using microevolution to explain the macroevolutionary observations for the evolution of sex*. In: *Macroevolution: Explanation, Interpretation and Evidence*, Volume 2, 279-299 (Springer).
- Hiltunen, T. & **L. Becks**, 2014. *Consumer co-evolution as an important component of the eco-evolutionary feedback*. **Nature Communications** 5, 5226, doi:10.1038/ncomms6226.
- Koch, H., Frickel, J. Valiadi, M. & **L. Becks**, 2014. *Why rapid, adaptive evolution matters for community dynamics*. **Frontiers in Ecology and Evolution** 2:17. doi: 10.3389/fevo.2014.00017/.
- Becks, L.** & A.F. Agrawal, 2013. *Higher rates of sex evolve under K-selection*. **Journal of Evolutionary Biology** 26, 900-905.
- Becks, L.** & H. Arndt, 2013. *Different types of synchrony in chaotic and cyclic communities*. **Nature Communications** 4, doi:10.1038/ncomms2355.
- Becks, L.** & A.F. Agrawal, 2012. *The evolution of sex is favoured during adaptation to new environments*. **PLoS Biology** 10, 1-11. (Recommended by Faculty 1000).
- Becks, L.**, S.P. Ellner, Jones, L. E. & N.G. Hairston Jr., 2012. *The functional genomics of an eco-evolutionary feedback loop: linking gene expression, trait evolution, and community dynamics*. **Ecology Letters** 15, 492-501. (Recommended by Faculty 1000).

Ellner, S.P. & **L. Becks**, 2011. *Rapid prey evolution in response to two predators*. **Theoretical Ecology** 4,133-152.

Becks, L. & A.F. Agrawal, 2011. *The effect of sex on the mean and variance of fitness in facultatively sexual rotifers*. **Journal of Evolutionary Biology**, 24, 656-664.

Becks, L., S.P. Ellner, Jones, L. E. & N.G. Hairston Jr., 2010. *Reduction of adaptive genetic diversity radically alters eco-evolutionary community dynamics*. **Ecology Letters**, 13, 989-997. (Recommended by Faculty 1000).

Becks, L. & A.F. Agrawal, 2010. *Higher rates of sex evolve in spatially heterogeneous environments*. **Nature**, 468, 89-92. (Recommended by Faculty 1000).

Jones, L. E., **Becks, L.**, Ellner, S.P., Hairston Jr. N.G., Yoshida,T. & G. Fussmann, 2009. *Rapid contemporary evolution and clonal food web dynamics*. **Phil. Trans. R. Soc. B.**, 364,1579-1591.

Becks, L. & H. Arndt, 2008. *Transitions from stable equilibria to chaos, and back, in an experimental food web*. **Ecology**, 89, 3222-3226.

Becks, L., Hilker, F., Malchow. H., Jürgens, K. & H. Arndt, 2005. *Experimental demonstration of chaos in a microbial food web*. **Nature**, 435, 1226-1229. (Recommended by Faculty 1000).

GRANTS AND AWARDS

Total: ~4,150,000 EUR

"Host virus coevolution – demography versus selection in the face of multiple stressors" German Research Foundation (DFG) **334,497 EUR**

"Indirect response to external drivers through trait variation in predator-prey systems" German Research Foundation (DFG), 2018 - 2021 **382,550 EUR**

Heisenberg Professorship from the German Research Foundation (DFG) at the Christian Albrechts University Kiel, 2017-2020) **373,000 EUR**

"Eco-evolutionary dynamics in evolving communities" German Research Foundation (DFG), 2017 - 2019 **233,650 EUR**

"Persistence mechanisms in a tripartite protist – giant virus – virophage system" (Betty and Gordon Moore Foundation) Investigator: L. Becks & M. Fischer (MPI Heidelberg), Jan. 2017 – Dec. 2108 part LB: **200,000 USD**

"Experimental Evolution and Community Dynamics - EECD 2016" (DFG, International Scientific Events) 2016 **10,200 EUR**

"Host virus coevolution – demography versus selection" Lead-Agency program" with SNF (P. Feulner EAWAG) as part of the Priority program by the German Research Foundation (DFG) SPP1819: "Rapid Evolution", Jan. 2016 - Dec. 2018 part LB: **322,328 EUR**

"Evolution in sexual and asexual populations" (Max-Planck Society), 2015 **80,000 EUR**

"Impact of the trade-offs on eco-evolutionary feedbacks in predator-prey systems" as part of the Priority program (DFG) SPP1704: "Flexibility matters: Interplay between trait diversity and ecological dynamics using aquatic communities as model systems (DynaTrait) "; Principal

Investigator: L. Becks & U. Gaedke (University of Potsdam,) Oct. 2014 - Oct. 2017

part LB: **187,805 EUR**

"Coevolving communities in spatially structured and heterogeneous environments" (DAAD/Academy of Finland); Principal Investigator: L. Becks & T. Hiltunen, Jan. 2014 - Dez. 2016

part LB: **10,963 EUR**

"Eco-evolutionary dynamics in complex systems" Emmy Noether Grant (DFG), Oct. 2011 - Oct. 2015

1,239,649 EUR

"The effects of sex on the coevolution and dynamics of a predator-prey system" (Volkswagen Foundation Postdoctoral fellowship), Oct. 2008 - Oct. 2011

193,000 EUR

"Studying complex dynamics: Contemporary rapid evolution in complex ecological communities" (James S. McDonnell Foundation); Principal Investigators: S. P. Ellner, N. G. Hairston Jr., L. Jones, L. Becks (all Cornell University, Ithaca), G. Fussmann (McGill University, Montreal), 2007 - 2011

449,459 USD

Others:

- Shortlisted for W3 position at University Mainz (2nd position; 2015).
- Shortlisted for professorship IST Austria (2016).
- Shortlisted for lecture at Queen Mary University London (2016).
- Young scientist award from the German Limnological Society (DGL), 2005.
- Excellent student scholarship for graduate studies from the University of Cologne (Germany).

INVITED PRESENTATIONS

I have been invited to present my research at internationally established conferences, symposia and advanced schools (**selected for the last 3 years, see below**).

- | | |
|------|---|
| 2018 | Sex uncovered: the evolutionary biology of reproductive systems, CONFERENCES JACQUES MONOD (France) |
| 2018 | Workshop "Experimental approaches to test for coevolution, Paris (France). |
| 2017 | Ringberg Symposium - Giant Viruses (Germany); Visions in Science, Berlin (Germany); Workshop "Origin of Biodiversity - Host-pathogen co-evolution in the genomic era" Gothenburg (Sweden). |
| 2016 | EVENT, Ghent (Belgium); ETH Zürich (Switzerland); IST Austria (Austria), Workshop "Experimental Evolution: Theory and Current Practices", Paris (France); Workshop "The genomic basis of eco-evolutionary change", Monte Veritas (Switzerland). |
| 2015 | European Phycological Congress, London (UK), German Ecological Society, Göttingen (Germany). |

SUPERVISION

Bachelor: Niklas Niehus (2015), Annika Friedrichsen (2016).

Master: Noemi Woltermann (2015), Sophia Wagner (2015), Aline Jeschkes (2016), Chris Gundlach (2016), Lasse Harder (2017), Ruben Hermann (ongoing).

PhD: Jens Frickel (defended 26.4.2016), Hanna Koch (defended 9.2016), Gökce Ayan (defended 11. 2017), Loukas Theodosiou (ongoing), Noemi Woltermann (currently on maternal leave). Elena Horas (ongoing).

Postdoc: Martha Valiadi (2.2013 -10.2013), Amrei Binzer (9.2013 - 9.2014), Heike Gruber *Aging and epigenetic modifications* (4.2014 - 1.2016), Vienna Kowallik (1.1.2016 - 31.1.2017), Ana del Arco: (1.4.2017-31.12.2018)

SCIENTIFIC SERVICES

- Faculty of the International Max Planck Research School Evolutionary Biology and member of several thesis committees
- Associate Editor Ecology Letters
- Associate Editor BMC Evolutionary Biology
- Associate Editor Frontiers in Ecology and Evolution
- Organization of the Workshop 'Bridging Theory and Experiments ~Evolving Communities' MPI Plön (2016)
- Organization of Mini-Symposium 'Experimental Evolution and Community Dynamics' University of Helsinki together with Dr. Teppo Hiltunen (2014, 2015) MPI Plön (2016)
- Organization of Symposium "Rapid evolution and population genetics" together with Dr. Teppo Hiltunen at the ESEB meeting in Lisboa (2013)
- Organization of Symposium "Ecology and Evolution of Sex" together with Hanna Koch at the ESEB meeting in Lausanne (2015)
- Organization of Graduate meeting of the German Zoological Society, MPI Plön (2017)
- Co-speaker of DFG priority program DynaTrait
- Co-organization of Kick-off meeting program DynaTrait (2014), annual Meeting (2015, 2016), international meeting (2017)
- Co-organization of Seminar series at the MPI Plön: Meselson Seminar on the Evolution of Sex
- Review for scientific journals: American Naturalist, BMC Biology, BMC Evolutionary Biology, Current Biology, Ecology, Ecology Letters, Evolution, Hydrobiologia, ISME Journal, Journal of Animal Ecology, Journal of Evolutionary Biology, Journal of Phycology, Limnology, Limnology and Oceanography, Molecular Ecology, Molecular Systems Biology, Nature Communications, PNAS, Trends in Ecology and Evolution
- Review or funding agencies: German Research Foundation (DFG), Swiss National Science Foundation (SNSF), Biotechnology and Biological Sciences Research Council (BBSRC), University of Bordeaux, University of Helsinki, FWO (Belgium), ERC.
- Outreach:
 - Darwin Day University of Kiel; Talk for 1200 high school students.
 - Public lectures in 2014, 2015 at the Max Planck Institute for Evolutionary Biology on the evolution of sex.
 - TV interview with public TV station (NDR) explaining the problem of the evolution of sex (2015).
- Scientific representative of the MPI Evolutionary Biology in the Max Planck Society.

CURRICULUM VITAE

1. CONTACT INFORMATION:

MALENA SABATINO, E-mail: malenasabat@gmail.com

Laboratorio de Artrópodos, Universidad Nacional de Mar del Plata.

Funes 3350 (7600) Mar del Plata, Buenos Aires, Argentina.

Tel. +54 223 4752426 int. 450

Scholar Google: <https://scholar.google.com.au/citations?user=I6BAhtQAAAAJ&hl=en>

ResearchGate: https://www.researchgate.net/profile/Malena_Sabatino

2. CURRENT POSITION:

- Researcher of CONICET (National Council of Scientific Research)

3. UNIVERSITY STUDIES:

- 2003 Graduated degree in Biological Science. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2012 Ph.D. in Biological Science. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.

4. RESEARCH INTERESTS:

My research interest is related to understand how the anthropogenic habitat alteration can affect the diversity and composition of the interactions assemblages, as well as web structure in plant-pollinator communities.

Moreover, I am interested in applying the ecological network approach as a tool for developing common strategies for biodiversity conservation and ecological restoration, landscape planning, and sustainable management of natural habitats.

5. GRANTS AND FELLOWSHIPS:

- CONICET Doctoral Grant (2007–2012) Laboratorio de Agroecología, INTA Balcarce, Buenos Aires / Laboratorio Ecotono, INIBIOMA, Río Negro. Argentina.
- FAPESP Postdoctoral Grant (2012) São Paulo State University (UNESP), Río Claro, São Pablo, Brasil.
- CONICET Postdoctoral Grant (2013–2015) Laboratorio Ecotono, UNCOMA / INTA Balcarce, Argentina.

- Argentinean National Agency of Scientific and Technological promotion (funding for young research) PICT-2015-1691 (2016-2018).

6. TEACHING ACTIVITIES:

- 2002–2003 Principles of Geology: Teacher assistant of practical course for graduation students. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2006 Extension course: "Plantas aromáticas, extracción de aceites esenciales y usos múltiples". Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2006–2007 Training extension course of Perito Apicultor. Teacher assistant. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2012 Chemistry for the introductory course: Teacher assistant for post-graduate students. Universidad Nacional del Comahue. San Carlos de Bariloche, Río Negro, Argentina.
- 2014 Vascular Plants. Teacher assistant for post-graduate students. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.
- 2017 Teacher assistant for post-graduate students course: "¿Abejas en crisis?: Una mirada interdisciplinaria sobre la actualidad Argentina y mundial". Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.

7. RESEARCH ADVISOR:

- MSc Student: Agustín Saez, co-supervised with Marcelo Aizen. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina (2009-2010).

8. RESEARCH COLLABORATION AND TRAINING:

- Dr. Jason Tylianakis group from June 2008 to January 2009. School of Biological Sciences, University of Canterbury, New Zealand.
- Dr. Marcelo Aizen group (2007-2012). Universidad Nacional del Comahue, San Carlos de Bariloche, Río Negro, Argentina.
- Dr. Nestor Maceira (2007-2018) EEA INTA Balcarce, Buenos Aires, Argentina.
- Dr. Martín Eguaras (2016-2018) Lab. Artrópodos. Universidad Nacional de Mar del Plata, Buenos Aires, Argentina.

9. PUBLICATIONS:

- SABATINO M., IURLINA M., EGUARAS M. & R. FRITZ, 2006. Tipificación Botánica y Caracterización de las Mielés del Sudeste de la Provincia de Buenos Aires. *Vida Apícola*, Revista de Apicultura N° 137. Editores Montagud. Barcelona. España.
- SABATINO M., N. MACEIRA & M. AIZEN, 2010. Direct effects of habitat area on interaction diversity in pollination webs. *Ecological Applications*, Vol 20(6), pp. 1491–1497.
- HERRERA L. & M.SABATINO, 2010. El valor de los pastizales serranos. *Visión Rural*. Año XVII N° 84, pp. 25-27.
- SÁEZ A., SABATINO M. & M. AIZEN, 2011. Interactive Effects of Large- and Small-Scale Sources of Pollinator Service in the Argentine Pampas. *PLoS One*, (7) 1.
- AIZEN M., SABATINO M. & J. TYLIANAKIS, 2012. Specialization and Rarity Predict Non-Random Loss of Interactions from Mutualist Networks. *Science* Vol. 335, pp 1486-1489.
- SÁEZ A., SABATINO M. & M.A. AIZEN (2014). La diversidad floral del borde afecta la riqueza y abundancia de visitantes florales nativos en cultivos de girasol. *Ecología Austral* 24: 94-102.
- CARSTENSEN D. W., SABATINO M., TRØJELSGAARD K. & L. P. C. MORELLATO (2014). Beta diversity of plant-pollinator networks and the turnover of pairwise interactions across space. *PLoS ONE* 9(11) e112903.
- CHALCOFF, V. R.; CAROLINA L. MORALES; MARCELO A. AIZEN; YAMILA SASAL; ADRIANA E. ROVERE; MALENA SABATINO; CAROLINA QUINTERO; MARIANA TADEY (2014). Interacciones planta-animal, la polinización. En: Raffaele E. de Torres Curth M., Morales C.L. & T. Kitzberger (Eds.), *Ecología e historia natural de la Patagonia Andina: un cuarto de siglo de investigación en biogeografía, ecología y conservación*. Ciudad Autónoma de Buenos Aires, Fundación de Historia Natural Félix de Azara.
- GILARRANZ L. J., SABATINO M., AIZEN M. A. & J. BASCOMPTE (2015). Hot-spots of mutualistic networks. *Journal of Animal Ecology*, Vol 84(2), 407-413.
- SABATINO M., N. MACEIRA & A. ROVERE (2015). Germinación de *Eryngium regnellii*: especie clave para la restauración ecológica de las interacciones planta-polinizador en la Pampa Austral (Buenos Aires, Argentina). *Revista Internacional de BOTÁNICA Experimental Phytón* Vol 84: 435-443.

- AIZEN, M. A., GLEISER, G., SABATINO, M., GILARRANZ, L. J., BASCOMPTE, J., & VERDÚ, M. (2016). The phylogenetic structure of plant-pollinator networks increases with habitat size and isolation. *Ecology letters*, 19(1), 29-36.
- CARSTENSEN D. W., SABATINO M. & L. P. C. MORELLATO (2016). Modularity, pollination systems, and interaction turnover in plant-pollinator networks across space. *Ecology* 97 (5), 1298-1306.
- SABATINO M. & N. MACEIRA (2016). El rol de las sierras en la conservación de la biodiversidad pampeana. *Visión Rural*. Año XXIII N° 112, pp. 46-48.
- HERRERA L., M. SABATINO, A. GASTÓN, S. SAURA (2017). Grassland connectivity explains entomophilous plant species assemblages in an agricultural landscape of the Pampa Region, Argentina. *Austral Ecology* 42: 486-496.
- HERRERA L., M. SABATINO, F. R. JAIMES & S. SAURA (2017). Landscape connectivity and the role of small habitat patches as stepping stones: An assessment of the grassland biome in South America. *Biodiversity and Conservation* 26: 3465.
- HERRERA L., F. R. JAIMES, M. SABATINO & S. L. POGGIO (2017). Una propuesta para valorar el estado de conservación de los bordes de caminos rurales en el sudeste bonaerense. *Ecología Austral* 27:404-414.
- SABATINO M., FARINA J. & MACEIRA N. (2017). Flores de las sierras de Tandilia: Guía para el reconocimiento de las plantas y sus visitantes florales. INTA Ediciones, Colección Recursos.

10. CONGRESS PARTICIPATION (Last 5 years):

- CARSTENSEN D.W., SABATINO M., MORELLATO L.P.C. Spatial turnover of pairwise plant-pollinator interactions. ATBS 2013. San José, Costa Rica.
- HERRERA L.P., JAIMES F., SABATINO M., POGGIO S. Una nota al margen: una propuesta para valorar el estado de conservación de los bordes de caminos en el Sudeste Bonaerense. XXVI Congreso Argentino de Ecología. Libro de resúmenes RAE, Noviembre 2014. Rio Negro, Argentina.
- JAIMES F., SABATINO M., HERRERA L. Mapeo de pastizales remanentes del Sistema de Tandilia (Buenos Aires): Un diagnóstico de la situación actual. XXVI Congreso Argentino de Ecología. Libro de resúmenes RAE, Noviembre 2014. Rio Negro, Argentina.
- ROVERE A.E., SABATINO M. Restauración ecológica de las interacciones planta-polinizador en la matriz agrícola Pampeana. IV Congreso Iberoamericano y del

Caribe de Restauración Ecológica. Libro de resúmenes SIACRE, Abril 2015. Buenos Aires, Argentina.

- ROVERE A.E., SABATINO M., BRAN D., FARINA J. Especies claves para la restauración de las interacciones planta-polinizador en un área degradada de la Estepa Patagónica. III° Taller regional sobre rehabilitación y restauración de la Diagonal Árida de la Argentina. Libro de resúmenes, Octubre 2016. Puerto Madryn, Chubut, Patagonia Argentina.
- SABATINO M., FARINA JUAN, MACEIRA N. Flores de las sierras de Tandilia: Guía para el reconocimiento de las plantas y sus visitantes florales. Comunicación en la 1º Reunión Científica del Programa Nacional de Recursos Naturales, Gestión Ambiental y Ecorregiones: aportes a la agroecología desde la biodiversidad, la gestión ambiental, el estudio del clima y el ordenamiento territorial. Libro de resúmenes, Agosto 2016, Buenos Aires, Argentina.

Matías Arim

Current affiliation: Department of Ecology and Evolution, University of República (Universidad de la República), Maldonado, Uruguay, VA 20000. Ph: (598) 98 119807

Research focus: the interaction between landscape structure, energy, and biodiversity, as determinants of metacommunity organization and function. I particularly attempt to connect theoretical and empirical studies.

A. Education:

The University of República, Uruguay	1999	B. Sc.	Biology
The University of República, Uruguay	2001	M.Sc.	Ecology
P. Catholic University of Chile, Chile	2005	PhD	Ecology
Center for Advanced Studies in Ecology and Biodiversity, Chile	2006	Postdoc	Ecology

B. Appointments:

2015-Present	Full Professor, Program for the Development of Basic Science, Uruguay
2005-2009	Assistant Professor, University of República, Uruguay
2009-Present	Associate Professor, University of República, Uruguay
2007-2011	CoPI, Center for Advanced Studies in Ecology and Biodiversity, CASEB, Chile (P. Catholic University of Chile).
2014-2018	Member of the Global Young Academy

C. Publications *Highlighted from 4 books chapters and 50 papers ISI*

1. Arim, M. & O. Barbosa. 2002. **Humped pattern of diversity: fact or artifact?** *Science*. 297:1763. <http://www.sciencemag.org/cgi/reprint/297/5588/1763a.pdf>
2. Arim, M. & P.A. Marquet. 2004. **Intraguild Predation: a significant and widespread interaction.** *Ecology Letters* 7: 557-564.
3. Arim M. & F. M. Jaksic. 2005. **Productivity and food web structure: association between productivity and link richness among top predators.** *Journal of Animal Ecology* 74: 31-40.
4. ¹Arim, M, S.R. Abades, P.E. Neill, M. Lima & P.A. Marquet. 2006. **Spread dynamics of invasive species.** *Proceeding of the National Academy of Science USA*. PNAS 103: 374-378.
5. Arim, M, Pablo A. Marquet & F.M. Jaksic. 2007. **On the relationship between productivity and food chain length at different ecological levels.** *The American Naturalist* 169: 62-72.
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7. Lafferty, K.D.; Allesina S.; **M. arim**; Briggs, C.J.; De Leo, G.; Dobson, A.P.; Dunne, J.A.; Johnson, P.T.J.; Kuris, A.M.; Marcogliese, D.J.; Martinez, N.D.; J. Memmott, J.; P.A. Marquet; J.P. Mclaughlin; E.A. Mordecai; M. Pascual; R. Poulin; D.W. Thieltges, D.W. 2008. **Parasites in food webs: the ultimate missing links.** *Ecology Letters* 11: 533-546.
8. Ramos-Jiliberto, R., A. Albornoz, F. Valdovinos, C. Smith-Ramírez, **M. Arim**, J. Armesto, P.A. Marquet. 2009. **Mutualistic networks, pollination web, degree distribution, nestedness, network dynamics.** *Oecologia*. 160: 697-706.
9. **Arim, M.**, S.R. Abades, G. Laufer, M. Loureiro & P.A. Marquet. 2010 **Food web structure and body size: trophic position and resource acquisition.** *Oikos* 119:147-153.
10. Toranza, C. & **M. Arim**. 2010. Cross-taxon congruence and environmental conditions. *BMC Ecology* 10:18.
11. Carranza, A. Defeo, O. & **Arim, M.** 2011. **Taxonomic relatedness and spatial structure of a shelf benthic gastropod assemblage.** *Diversity and Distribution* 17: 25-34.
12. **Arim, M.**, M. Berazategui, J. M. Barreneche, L. Ziegler, M. Zaruki, and S. R. Abades. 2011. **Determinants of density–body size scaling within food webs and tools for their detection.** *Advances in Ecological Research* 45:1-40.
13. Borthagaray, A.I., **Arim, M.** and Marquet, P.M. 2012. **Connecting landscape structure and patterns in body size distribution.** *Oikos* DOI: 10.1111/j.1600-0706.2011.19548.x).
14. Garay-Narvaez, L., **Arim, M.**, Flores, J.D., Ramos-Jiliberto, R. 2013. **The more polluted the environment, the more important biodiversity is for food web stability.** *Oikos*, 122: 1247-1253.
15. Ziegler L., Berazategui M. & **Arim M.** 2014 **Discontinuities and alternative scalings in the density–mass relationship of anuran larvae.** *Hydrobiologia*. 723:123-129. doi: 10.1007/s10750-013- 1553-2.
16. Borthagaray, A.I., Barreneche, J.M. Abades, S.R., & **Arim, M.** 2014 **Modularity along organism dispersal gradients challenges a prevailing view of abrupt transitions in animal landscape perception.** *Ecography*. 37(6):564-571. DOI: 10.1111/j.1600-0587.2013.00366.x
17. Garay-Narvaez, L., Flores, J.D., **Arim, M.**, Ramos-Jiliberto, R. 2014. **Food web modularity and biodiversity promote species persistence in polluted environments.** *Oikos*, 123: 583-588. DOI: 10.1111/j.1600-0706.2013.00764.x
18. Borthagaray, A.I., **Arim, M.**, Marquet, P.A. 2014. **Inferring species roles in metacommunity structure from species co-occurrence networks.** *Proceedings of the Royal Society B*. 281:. DOI: <http://dx.doi.org/10.1098/rspb.2014.1425>
19. Segura, A., Calliari, D. Kruk, C., Fort, H., Izaguirre, I., Saad, J., **Arim, M.** 2015 **Metabolic dependence of phytoplankton species richness.** *Global Ecology and Biogeography* 24: 472-482. DOI: 10.1111/geb.12258
20. Borthagaray, A.I., Berazategui, M., & **Arim, M.** 2015. **Disentangling the effects of local and regional processes on biodiversity patterns through taxon-contingent metacommunity network analysis.** *Oikos*. 124: 1383-1390. DOI: 10.1111/oik.01317
21. Borthagaray, A.I., Pinelli, V., Berazategui, M., Rodríguez-Tricott, L. and **M. Arim**. 2015. **Effects of metacommunity network on local communities structure: from theoretical predictions to empirical evaluations.** En *Aquatic Functional Biodiversity* A. Belgrano, G. Woodward and U. Jacob (Eds). Elsevier
22. **Arim, M.**, Borthagaray, A.I., Giacomini, H.C. 2016. **Energetic constrains to food chain lenght in a metacommunity framework.** *Candaina Journal of Fisheries and Aquatic Sciences*. 73(4): 685-692. DOI: 10.1139/cjfas-2015-0156
23. Segura, A. M., Fariña, R. A., & **Arim, M.** 2016. **Exceptional body sizes but typical**

- trophic structure in a Pleistocene food web.** *Biology Letters*, 12(5), 20160228
24. Segura, A. M., Calliari, D., Lan, B. L., Fort, H., Widdicombe, C. E., Harmer, R., & Arim, M. 2017. **Community fluctuations and local extinction in a planktonic food web.** *Ecology Letters*, 20(4), 471-476.
 25. Canavero, A., Arim, M., Pérez, F., Jaksic, F. M., & Marquet, P. A. (2018). **A metabolic view of amphibian local community structure: the role of activation energy.** *Ecography*.41: 388-400.
 26. Borthagaray, A.I., Soutullo, A., Carranza, A. Arim, M. 2018. A modularity-based approach for identifying biodiversity management units. *Revista Chilena de Historia Natural*. 91: 2.

D. Synergistic Activities:

D.1. Reviewer for Journals:

- | | |
|---|---|
| 1. Acta Oecologica | 20. Frontiers in Ecology |
| 2. Aquatic Ecology | 21. Functional Ecology |
| 3. Biological Invasions | 22. F1000Research |
| 4. Biological Research | 23. Gayana |
| 5. BMC Ecology | 24. Global Ecology and Biography |
| 6. Boletín de la Sociedad Zoológica del Uruguay | 25. Hydrobiologia |
| 7. Canadian Journal of Fisheries and Aquatic Sciences | 26. Ibis |
| 8. Canadian J. of Zoology | 27. Journal of Animal Ecology |
| 9. Climate Research | 28. Journal of Arid Environments |
| 10. Conservation Biology | 29. Journal of Biogeography |
| 11. Diversity and Distribution | 30. Journal of Ecology |
| 12. Ecography | 31. Journal of Mammalogy |
| 13. Ecological Entomology | 32. Journal of Zoology |
| 14. Ecological Modelling | 33. Marine Ecology Progress Series |
| 15. Ecology | 34. Marine Biodiversity |
| 16. Ecology Letters | 35. Neotropical Ichthyology |
| 17. Ecology and Evolution | 36. Oecologia |
| 18. Ecosistemas | 37. Oikos |
| 19. Evolutionary Ecology Research | 38. Revista Chilena de Historia Natural |
| | 39. Theoretical Ecology |
| | 40. Wildlife Research |

D.2. Reviewer of proposals

National Science Organization of Argentina, Austria, Chile, Colombia, Panamá, Uruguay, and for the European Research Council.

E. Advisor

Advisor of 16 undergraduate, 16 master, and 8 PhD students in programs of Argentina, Chile, Spain, and Uruguay. Now advisor of 3 Master, and 4 PhD students.

F. Principal Investigator in Grants *(highlighted from a total of 10)*

Arim, M. 2008-2018. Programs for University Development. PI of the Program "Development of Research and Human Resources on Biodiversity"

Arim, M. 2013-2015. "Metabolism, randomness, scale, and entropy: empirical evaluation of new theories in community Ecology" (Fondo Clemente Estable 2011, Metabolismo, azar, escalas y entropía: evaluación empírica de nuevas teorías en ecología de comunidades).

Arim, M. 2016-2018. "Effect of metacommunity network on local communities: from theoretical predictions to empirical evaluations" (Fondo Clemente Estable 2014, Efecto de la red metacomunitaria en las comunidades locales: desde las predicciones teóricas a la evaluación empírica). 2016-2017.

G. Responsible of undergraduate and graduate courses

Theory and methods in metacommunity ecology. Course directed to graduate students, organized with Dr. Ana Borthagaray in Argentina and Uruguay (since 2017).

Community Ecology. Course directed to graduate students (since 2012).

Ecology. Course directed to undergraduate students (since 2013).

Theoretical Ecology. Course directed to undergraduate students (since 2013).

F. Presentations

1. **2018. Arim M., Pinelli V., Rodríguez L., Ortiz E., Illarze M., Fagúndez C., Borthagaray A. Ponds size and isolation determining the relative role of niche versus neutral assembly of plants communities.** Nordic Society OIKOS, Trondheim, Norway.
2. **2017. Arim, M. Gastropods in food webs from general predictions to idiosyncratic patterns.** X Latin-American Congress of Malacology, Maldonado, Uruguay. Plenary Conference.
3. **2016. Arim, M. Metacommunities and Ecological Theory Integration.** Network for Ecological Theory Integration (NETI), Satan Fe Institute, USA.
4. **2015. Arim, M. The importance of take a look outside the pond to understand pond ecology.** F. de Ciencias, Universidad de la República, Uruguay. II Simposio de Peces Anuales. Plenary Conference.
5. **2015. Arim, M. Networks impacting networks: the interrelationship between metacommunities and food webs.** XXII Reunión Anual de la Sociedad de Ecología de Chile. Plenary Conference.

Full name: Ruben Hüttel Heleno
Current position: Auxiliary researcher (Investigador FCT)
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Homepage: <http://cfe.uc.pt/#/profile/members/14>
Google scholar: <https://scholar.google.co.uk/citations?user=PR5Wrr4AAAAJ&hl=pt-PT>
ResearcherID: <http://www.researcherid.com/rid/A-5778-2011>
citations: Web of Science= 591; Google Scholar= 101
h-index: Web of Science= 15 ; Google Scholar= 19

Higher education

- ♦ **2005-2009** PhD in Community Ecology: “The impact of alien plants on native biota: a food Web approach”, School of Biological Sciences, University of Bristol, UK. (“Doutoramento” registered at the University of Coimbra in July 2013). No corrections.
- ♦ **1999-2004** Biology Degree, specialization in Animal Ecology, (average grade 16/20), University of Coimbra, Portugal.

Selected publications ISI (5 most relevant highlighted)

- ⁵¹ J Costa, J Ramos, L Silva, S Timóteo, P Andrade, P Araújo, C Carneiro, E Correia, P Cortez, M Felgueiras, C Godinho, R Lopes, C Matos, A Norte, P Pereira, A Rosa, R Heleno *in press* Rewiring of experimentally disturbed seed dispersal networks might lead to unexpected network configurations. **Basic and Applied Ecology** (I.F.=2.1; Q1)
- ⁵⁰ Correia M, R Heleno, P Vargas, S Rodriguez-Echeverría *in press* Should I stay or should I go? Mycorrhizal plants are more likely to invest in long-distance seed dispersal than non-mycorrhizal plants. **Ecology Letters** doi:10.1111/ele.12936 (I.F.=9.45; Q1)
- ⁴⁹ Olesen J, C Damgaard, F Fuster, R Heleno, M Nogales, B Rumeu, K Trøjelsgaard, P Vargas, A Traveset 2018 Disclosing the double mutualist role of birds on Galápagos. **Scientific Reports** 8:57 doi:10.1038/s41598-017-17592-8 (I.F.=4.85; Q1)

 ⁴⁸ Timóteo S, M Correia, S Rodríguez-Echeverría, H Freitas, R Heleno 2018 Multilayer networks reveal the spatial structure of seed-dispersal interactions across the Great Rift landscapes. **Nature Communications** 9:140 doi:10.1038/s41467-017-02658-y (I.F.=12.12; Q1)

- ⁴⁷ Vargas P, M Fernández-Mazuecos, R Heleno 2018 Phylogenetic evidence for Miocene origin of Mediterranean lineages: species diversity, reproductive traits and geography. **Plant Biology** 20(S1):157-165 doi:10.1111/plb.12626 (I.F.=2.10; Q1)
- ⁴⁵ Arjona Y, M Nogales, R Heleno, P Vargas 2018 Long-distance dispersal syndromes matter: diaspore-trait effect on shaping plant distribution across the Canary Islands. **Ecography** doi:10.1111/ecog.02624 (I.F.=5.41; Q1)

 ⁴⁴ Rumeu B, M Devoto, A Traveset, J Olesen, P Vargas, M Nogales, R Heleno 2017 Predicting the consequences of disperser extinction: richness matters the most when abundance is low. **Functional Ecology** 31(10): 1910-1920, doi:10.1111/1365-2435.12897 (I.F.=5.1; Q1)

- ⁴³ Nogales M, A González-Castro, B Rumeu, A Traveset, P Vargas, P Jaramillo, J Olesen, R Heleno 2017 Contribution by vertebrates to seed dispersal effectiveness in the Galápagos islands: a community-wide approach. **Ecology** 98(8) 2049–2058, doi:10.1002/ecy.1816 (I.F.=4.7; Q1)
- ⁴² Patiño J, R Whittaker, P Borges, J Fernández-Palacios, C Ah-Peng, M Araújo, S Ávila, P Cardoso, J Cornuault, E de Boer, L de Nascimento, A Gil, A González-Castro, D Gruner, R Heleno, J Hortal, J Illera, C Kaiser-Bunbury, T Matthews, A Papadopolou, N Pettorelli, J Price, A Santos, M Steinbauer, K Triantis, L Valente, P Vargas, P Weigelt, B Emerson, 2017. A roadmap for island biology: 50 fundamental questions after 50 years of The Theory of Island Biogeography. **Journal of Biogeography** 44(1):963-983, doi:10.1111/jbi.12986 (I.F.=4.00; Q1)

 ⁴¹ Heleno R 2017 Birds make the World go round. **Trends in Ecology and Evolution** 32(3):154-155, doi:10.1016/j.tree.2017.01.001 (I.F.=16.74; Q1)

- ⁴⁰ López-Núñez F, R Heleno, S Ribeiro, H Marchante, E Marchante 2017. Four-trophic level food webs reveal the cascading impacts of an invasive plant targeted for biocontrol. **Ecology** 98(3):782-793, doi:10.1002/ecy.1701 (I.F.=4.7; Q1)
- ³⁹ Guzmán B, R Heleno, M Nogales, W Simbaña, A Traveset, P Vargas 2017 Evolutionary history of the endangered shrub snapdragon (*Galvezia leucantha*) of the Galápagos Islands. **Diversity and Distributions** 23(3):247-260, doi:10.1111/ddi.12521 (I.F.=4.6; Q1)
- ³⁸ da Silva L, J Ramos, A Coutinho, P Tenreiro, R Heleno 2017 Flower visitation by European birds offers the first evidence of interaction release in continents. **Journal of Biogeography** 44(3):687-695, doi:10.1111/jbi.12915 (I.F.=4.00; Q1)
- ³⁷ Rodríguez-Echeverría S, H Teixeira, M Correia, S Timóteo, R Heleno, M Öpik, M Moora 2017 Arbuscular mycorrhizal fungi communities from tropical Africa reveal strong ecological structure. **New Phytologist** 213(1):380–390, doi:10.1111/nph.14122 (I.F.=7.21; Q1)

- ³⁶ Correia M, S Timóteo, S Rodríguez-Echeverría, A Mazars-Simón, R Heleno 2017 The refaunation of Gorongosa National Park, Mozambique and the reinstatement of the seed dispersal function. **Conservation Biology** 31(1):76-85, doi:10.1111/cobi.12782 (I.F.=4.27; Q1)
 - ³⁴ da Silva L, A Coutinho, R Heleno, P Tenreiro, J Ramos 2016 Dispersal of fungi spores by non-specialized flower-visiting birds. **Journal of Avian Biology** 47(3):438–442, doi:10.1111/jav.00806 (I.F.=1.97; Q1)
 - ³³ Traveset A, M Nogales, P Vargas, J Olesen, P Jaramillo, R Heleno 2016 The Galápagos land iguana (*Conolophus subcristatus*) as a seed disperser. **Integrative Zoology** 11(3):207-213, doi:10.1111/1749-4877.12187 (I.F.=1.9; Q1)
 - ³² Costa J, L da Silva, J Ramos, R Heleno 2016 Sampling completeness in seed dispersal networks: When enough is enough. **Basic and Applied Ecology** 17(2):155-164, doi:10.1016/j.baae.2015.09.008 (I.F.=2.4; Q1)
 - ³¹ Traveset A, C Tur, K Trøjelsgaard, R Heleno, R Castro-Urgal, J Olesen 2016 Global patterns of mainland and insular pollination networks. **Global Ecology and Biogeography** 25(7):880-890, doi:10.1111/geb.12362 (I.F.=7.26; Q1)
 - ³⁰ Nogales M, R Heleno, B Rumeu, A González-Castro, A Traveset, P Vargas, J Olesen 2016 Seed-dispersal networks on the Canaries and the Galápagos archipelagos: Interaction modules as biogeographical entities. **Global Ecology and Biogeography** 25(7):912-922, doi:10.1111/geb.12315 (I.F.=7.26; Q1)
 - ²⁹ Esteves C, J Costa, P Vargas, H Freitas, R Heleno 2015 On the limited potential of Azorean fleshy fruits for oceanic dispersal. **PLoS One** 10(10):e0138882, doi:10.1371/journal.pone.0138882 (I.F.=3.2; Q1)
 - ²⁸ Traveset A, S Chamorro, J Olesen, R Heleno 2015 Space, Time and Aliens: charting the dynamic structure of Galápagos pollination networks. **AoB Plants** 7:plv068, doi:10.1093/aobpla/plv068 (I.F.=2.27; Q2)
 - ²⁷ Vargas P, Y Arjona, M Nogales, R Heleno 2015 Long-distance dispersal to oceanic islands: success of plants with multiple diaspore specializations. **AoB Plants** 7:plv073, doi:10.1093/aobpla/plv073 (I.F.=2.27; Q2)
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|---|---|
|  | ²⁶ Traveset A, J Olesen, M Nogales, P Vargas, P Jaramillo, E Antolín, M Trigo, R <u>Heleno</u> 2015 Bird-flower visitation networks in the Galápagos unveil a widespread interaction release. Nature Communications 6:6376, doi:10.1038/ncomms7376 (I.F.=10.74; Q1) |
|  | ²⁴ <u>Heleno</u> R, P Vargas 2015 How do islands become green? Global Ecology and Biogeography 24(5):518-526, doi:10.1111/geb.12273 (I.F.=7.26; Q1) |
- ²³ da Silva L, J Ramos, J Olesen, A Traveset, R Heleno 2014 Flower visitation by birds in Europe **OIKOS** 123:1377-1383, doi:10.1111/oik.01347 (I.F.=3.56; Q1)
 - ¹⁹ Heleno R, C Garcia, P Jordano, A Traveset, JM Gómez, N Blüthgen, J Memmott, M Moora, J Cerdeira, S Rodríguez-Echeverría, H Freitas, J Olesen 2014 Ecological networks: Delving into the architecture of biodiversity. **Biology Letters** 10(1):20131000, doi:10.1098/rsbl.2013.1000 (I.F.=3.53; Q1)
 - ¹⁸ Costa J, J Ramos, L da Silva, S Timóteo, P Araújo, M Felgueiras, A Rosa, C Matos, P Encarnação, P Tenreiro, R Heleno 2014 Endozoochory largely outweighs epizoochory in migrating passerines. **Journal of Avian Biology** 45:59-64, doi:10.1111/j.1600-048X.2013.00271.x (I.F.=2.35; Q1)
 - ¹⁷ Cruz J, J Ramos, L da Silva, P Tenreiro, R Heleno 2013 Seed dispersal networks in an urban novel ecosystem. **European Journal of Forest Research** 132(5):887-897, doi:10.1007/s10342-013-0722-1 (I.F.=1.8; Q1)
 - ¹⁶ Traveset A, R Heleno, S Chamorro, P Vargas, C McMullen, R Castro-Urgal, M Nogales, H Herrera, J Olesen 2013 Invaders of pollination networks in the Galápagos Islands: emergence of novel communities. **Proceedings of the Royal Society - B** 280(1758):2012-3040, doi:10.1098/rspb.2012.3040 (I.F.=5.68; Q1)
 - ¹⁵ Heleno R, J Olesen, M Nogales, P Vargas, A Traveset 2013 Seed dispersal networks in the Galápagos and the consequences of alien plant invasions. **Proceedings of the Royal Society - B** 280(1750):2012-2112, doi:10.1098/rspb.2012.2112 (I.F.= 5.68; Q1)
 - ¹⁴ Heleno R, J Ramos, J Memmott 2013 Integration of exotic seeds into an Azorean seed dispersal network. **Biological Invasions** 15(5):1143-1154, doi:10.1007/s10530-012-0357-z (I.F.=2.8; Q1)
 - ¹² Nogales M, R Heleno, A Traveset, P Vargas 2012 Evidence for overlooked mechanisms of long-distance seed dispersal to and between oceanic islands. **New Phytologist** 194(2):313-317, doi:10.1111/j.1469-8137.2011.04051.x (I.F.=6.37; Q1)
 - ¹¹ Heleno R, M Devoto, M Pocock 2012 Connectance of ecological networks and conservation value. **Ecological Indicators** 14(1):7-10, doi:10.1016/j.ecolind.2011.06.032 (I.F.=3.67; Q1)
 - ¹⁰ Vargas P, R Heleno, M Nogales, A Traveset 2012 Colonization of the Galápagos Islands by plants with no specific syndromes for long-distance dispersal: a new perspective. **Ecography** 35(1):33-43, doi:10.1111/j.1600-0587.2011.06980.x (I.F.=4.02; Q1)
 - ⁹ Heleno R, S Blake, P Jaramillo, A Traveset, P Vargas, M Nogales 2011 Frugivory and seed dispersal in the Galápagos: What is the state of the art? **Integrative Zoology** 6 (2): 110-119, doi:10.1111/j.1749-4877.2011.00236.x (I.F.=1.42; Q1)
 - ⁸ Heleno R, G Ross, A Everard, J Memmott, J Ramos 2011 On the role of avian seed predators as seed dispersers. **Ibis** 153 (1): 199-203, doi:10.1111/j.1474-919X.2010.01088.x (I.F.=1.92; Q1)
 - ⁵ Ceia R, H Sampaio, S Parejo, R Heleno, M Arosa, J Ramos, G Hilton 2010 Throwing the baby out with the bathwater: does laurel forest restoration remove a critical winter food supply for the critically endangered Azores bullfinch? **Biological Invasions** 13(1):93-104, doi:10.1007/s10530-010-9792-x (I.F.=2.8; Q1)

- ⁴ Heleno R, I Lacerda, J Ramos, J Memmott 2010 Evaluation of restoration effectiveness: community response to the removal of alien plants. **Ecological Applications** 20(5):1191-1203, doi:10.1890/09-1384.1 (I.F.=4.21; Q1)
- ¹ Heleno R, R Ceia, J Ramos, J Memmott 2009 The effect of alien plants on insect abundance and biomass: a food web approach. **Conservation Biology** 23(2):410-419, doi:10.1111/j.1523-1739.2008.01129.x (I.F.=5.16;Q1)

Books and chapters

- ⁷ Escribano-Avila G, C Lara-Romero, R Heleno, A Traveset 2018 Tropical Seed dispersal Networks: emerging patterns, biases and keystone species traits. *in* Ecological Networks in the Tropics. W Dattilo and V Rico-Gray (eds), Springer, 93-110 ISBN-13:9783319682273.
- ⁶ Traveset A, R Heleno, M Nogales 2014 The ecology of seed dispersal. *in* Seeds: The ecology of regeneration in plant communities. R Gallagher (ed), Oxon, UK, CABI International, 62-93.
- ⁵ Gardener M, M Trueman, C Buddenhagen, R Atkinson, R Heleno, H Jäger, A Tye 2013 A pragmatic approach to management of plant invasions in Galapagos. *in* Plant Invasions in Protected Areas: Patterns, Problems and Challenges, Invading Nature - Springer Series in Invasion Ecology 7. C Llewellyn, D Richardson, P Pyšek, P Genovesi (eds) Dordrecht, Springer Science+Business Media, 349-374.
- ⁴ Jaramillo P, R Heleno 2012 Seeds of the Galápagos Islands. Puerto Ayora, Galapagos, Charles Darwin Foundation.
- ³ Borges P et al. 2010 List of Arthropods (Arthropoda) of the Azores. *in* P Borges et al. (ed). A list of the terrestrial and marine biota from the Azores. Principia Editora, Cascais.
- ² Several authors 2008 Novo guia das Aves Nidificantes de Portugal, edited by the Sociedade Portuguesa para o Estudo das Aves and Instituto de Conservação da Natureza e das Florestas
- ¹ Memmott J, R Gibson, L Carvalheiro, K Henson, R Heleno, M Lopezaraiza, S Pearce 2007 The conservation of ecological interactions. *in* A Stewart, T New, O Lewis (eds). Insect Conservation Biology. The Royal Entomological Society, London.

Selected oral presentation (invited plenary/keynote talks highlighted)

⁵³	<u>Heleno</u> R 2017 On the road to a novel natural world: How alien species change the structure and function of ecological networks. 14 th EMAP - Ecology and Management of Alien Plant Invasions, Lisbon, Portugal – INVITED PLENARY KEYNOTE
⁵¹	<u>Heleno</u> R 2017 Non-standard endozoochory and epizoochory by small passerines in the Azores, Galapagos and mainland Portugal. Workshop “Beyond frugivory and scatterhoarding” EBD-CSIC, Sevilla, Spain – INVITED
⁴⁹	<u>Heleno</u> R 2016 Macroecological patterns of species interaction networks: community assembly, “evolution” and disassembly, Trends in Biodiversity and Evolution – TiBE2016, CIBIO – Universidade do Porto, Oporto, Portugal – INVITED KEYNOTE
⁴⁸	<u>Heleno</u> R, A Coelho, F Mendes, R de Lima 2016 The impact of introduced mammals on the seed-dispersal network of São Tomé island, Island Ecology Symposium, University of Seychelles, Mahé, Seychelles – INVITED
⁴³	<u>Heleno</u> R 2016 The role of birds on ecosystem functions: a network approach, VI Iberian Ornithological Congress – SPEA & SEO, Vila Real, Portugal – INVITED PLENARY KEYNOTE
³⁷	<u>Heleno</u> R, Y Arjona, M Nogales, P Vargas 2015 Do dispersal syndromes matter? Inter-island plant dispersal across the Azores, Canaries, Galápagos and Cape Verde, European Ecological Federation meeting, Rome, Italy – INVITED
³³	<u>Heleno</u> R 2015 Interaction modules: disentangling the complexity of biological communities, 4 th Iberian Ecological Congress, Coimbra, Portugal – INVITED PLENARY KEYNOTE
²⁴	<u>Heleno</u> R 2014 Birds in species-interaction networks: where do we stand and where can we go from here?, British Ornithological Union one-day meeting: Birds in the entangled bank, Peterborough, UK – INVITED KEYNOTE
⁵	<u>Heleno</u> R 2008 Seed Dispersal Networks: Are Predators That Irrelevant for Seed Dispersal?, EGI Student Conference, University of Oxford, Oxford , UK – HIGH HONOUR MENTION FOR BEST TALK

Projects (raised >2.2M€ in funding)

- ♦ **International funding as principal investigator: 2013-2016 “SEEDS** - Closing the seed dispersal loop: How does seed dispersal affect plant population structure at the global, regional and local scales?” Centre for Functional Ecology, University of Coimbra. Marie Curie Career Integration Grant, European Union (FP7-PEOPLE-2012-CIG-321794) 100.000€.
- ♦ **International funding as team member: 2018-2020 “ISLET-FOODWEBS** - Effects of Global Change on small island trophic meta-networks” PI: Anna Traveset, Institut Mediterrani d’Estudis Avançats (CSIC-UIB), Spain. Spanish Ministerio de Economía y Competitividad (PE2017: CGL2017-88122-P) 196.000€.
- ♦ **International funding as team member: 2017-2020 “Spatio-temporal responses of ecosystem functions to ecological restoration”** PI: Christopher Kaiser-Bunbury, Darmstadt Technical University; Deutsche Gesellschaft für Finanzwirtschaft (DGF 2017 KA3349/2-2) 436.300€.
- ♦ **International funding as team member: 2016-2018 “COLONIZATION** - Características dispersivas: reconstrucción de la colonización de especies vegetales entre islas oceánicas” PI: Pablo Vargas, Real Jardín Botánico de Madrid (CSIC-RJB). Spanish Ministerio de Economía y Competitividad (PN2014: SPID201500X067883IV0) 168.205€.

- ♦ **International funding as team member: 2015-2017 “VERMUTIS** - Importance of single and double plant-vertebrate mutualisms on islands: Double benefits, double risks?” PI: Anna Traveset, Institut Mediterrani d’Estudis Avançats (CSIC-UIB), Spain. Spanish Ministerio de Economía y Competitividad (PE2013: CGL2013-44386-P) 183.000€.
- ♦ **International funding as team member: 2013-2016 “OCEIS** -The colonization history in the Galapagos islands by plants: floristic, diaspore and phylogenetic analysis”. PI: Pablo Vargas, Real Jardín Botánico de Madrid (CSIC-RJB). Spanish Ministerio de Economía y Competitividad (PN2012: CGL2012-38624-C02-01) 140.000€.
- ♦ **International funding as team member: 2010-2012 “REDGAL** - Redes mutualistas en las islas Galápagos: Impactos directos e indirectos de especies invasoras sobre plantas amenazadas” PI: Anna Traveset, Institut Mediterrani d’Estudis Avançats (CSIC-UIB), Spain. Fundación BBVA 199.166€.
- ♦ **International funding as team member: 2016 “Alien hitchhiker’s: The role of avian seed dispersal to São Tomé forest dynamics”**. PI: Ricardo de Lima, Centre for Ecology, Evolution and Environmental Changes, Universidade de Lisboa. The Rufford Foundation 12.517€.
- ♦ **National funding as principal investigator: 2014-2018 “SEEDS** - Closing the seed dispersal loop: How does seed dispersal affect plant population structure at the global, regional and local scales?” Centre for Functional Ecology, University of Coimbra. Fundação para Ciência e a Tecnologia (IF/00441/2013) 50.000€.
- ♦ **National funding as team member: 2018-2020 “BatPine** - Promoting the resilience of ecosystem services to climate change: a case study with pine plantations, pine processionary moth and bats?” PI: Hugo Rebelo, ICETA – Universidade do Porto. P2020 (31731) 223.869€.
- ♦ **National funding as team member: 2016-2019 “INVADER** - Ferramentas INoVadoras para Detectar Espécies invasoras e agentes de controlo biológico”. PI: Elizabete Marchante, Centre for Functional Ecology, University of Coimbra. Fundação para Ciência e Tecnologia (PTDC/AAG-REC/4896/2014) 199.922€.
- ♦ **National funding as team member: 2013-2015 “GORONGOSA** - Exploring biodiversity and ecological networks for forest conservation and restoration in Gorongosa National Park – Mozambique” PI: Susana Rodriguez-Echeverria, Centre for Functional Ecology, University of Coimbra. Fundação para Ciência e Tecnologia (PTDC/BIA-BIC/4019/2012) 180.972€.
- ♦ **National funding as team member: 2013-2015 “INVADER-B** - Invasive plant species management in Portugal: from early detection to remote sensing and biocontrol of *Acacia longifolia*” PI: Elizabete Marchante, Centre for Functional Ecology, University of Coimbra. Fundação para Ciência e Tecnologia (PTDC/AAG-REC/4607/2012) 199.983€.

Post-graduate supervision

- ♦ Post-Docs: Patricia Mateo-Tomás; Beatriz Rumeu Ruiz; Sérgio Antunes Timóteo
 - ♦ PhD students: Alba Costa Lorenzo; Helena Raposeira; Sasha Vasconcelos; Francisco Lopez-Nuñez; José Miguel Costa; Marta Correia; Luis Pascoal da Silva;
 - ♦ MSc students: Filipa Mendes; Ana Coelho; Mauro Nereu; Mariana Valente; Sérgio Ribeiro; Sanjay Sonet; Jioana Cruz.
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Outreach

Television

- 2018** news piece on the TV program “Olhar o Mundo” broadcasted by RTP3 and RTP Africa – 5 February.
- 2017** Invited scientist on the TV program “Sociedade Civil” broadcasted by RTP2 and RTP Internacional – 6 June.
- 2015** Invited scientist on the TV program “Mentes que Brilham” broadcasted by Porto Canal – 14 April.
- 2014** Communication of the Gorongosa project on the TV show “Praça da Alegria” broadcasted by RTP1 – 2 February.
- 2002** Participation on the TV program “Olhares Cruzados” broadcasted by RTP 2 – 26 November.

Radio

- 2017** Participation in the series of science outreach podcasts “Todas as plantas têm nome” broadcasted by the Rádio Universidade de Coimbra and published in the journal Diário de Coimbra.
 - 2015** Invited scientist on the radio program “Dias do Futuro” broadcasted by the national station Antena1 – 13 March.
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Scientific & Academic management duties

- Member of the scientific committee (“Comissão Científica”) of the Department of Life Sciences, Faculty of Sciences and Technology - University of Coimbra (2017 – present).
- Representative of doctorate members in the direction board of the R&D unit Centre for Functional Ecology – University of Coimbra (2016 – present).

Editorial boards

- Associate editor of the journal “Functional Ecology”, British Ecological Society (2017 – present).
- Subject editor of the journal “Web Ecology”, European Ecological Federation (2014 – present)

Academic juries

- PhD jury – Rocio Castro-Urgal, Sergio Chozas Vinuesa; Cristina Tur Espinosa; Danny Rojas Martí; Beatriz Rumeu Ruiz;

Steering committees & Project evaluation boards

External evaluator for the Spanish call for “Unique scientific and technical infrastructures” 2018
Evaluator for the Marie Skłodowska-Curie Individual Fellowships Europe (H2020-MSCA) 2016, 2017
External project evaluator for the Netherlands Organisation for Scientific Research (NOW) 2016
External project evaluator for the French Agence Nationale de la Recherche (ANR) 2014, 2015
External project evaluator for the Spanish Ministerio de Economía y Competitividad, 2013
IUCN Red List Authority for Birds and Species Survival Commission – Expert member (2012-present).

Organization of scientific meetings

2019 “X Portuguese Ornithological Congress - SPEA”. 1-5 March, Peniche, Portugal.
2019 “First Iberian Ecological Society Meeting CSIE2019”. 4-7 February, Barcelona, Spain.
2016 “Network approaches to island biology”, 2nd Island Biology meeting. 18-22 July, Terceira – Azores, Portugal.
2015 4th Iberian Ecological Meeting: “Ecology and the Societal challenges”. 16-19 June, Coimbra – Portugal.
2014 British Ornithologists Union meeting: “Birds in the entangled bank: advances in food-web theory and practice”. 19 November Peterborough – UK.
2013 1st International Symposium on “Ecological Networks: Delving into the architecture of biodiversity”. 23-25 October, Coimbra – Portugal.

Teaching (157 hours on 8 courses):

- ♦ **From 2016 to 2017** Advanced class on “Ecological networks” integrated in the “Lab and field course II” of the European Master in Applied Ecology, Department of Life Sciences, University of Coimbra – Lousã (2x8h). Coord.: José Paulo Sousa
- ♦ **From 2015 to 2017** Invited lecturer in the advanced course on “Tropical Ecology” of the master in Conservation Biology, Faculty of Sciences, University of Lisbon (3x1h). Coord.: Jorge Palmeirim.
- ♦ **From 2015 to 2017** Invited lecturer in classes of Bioinformatics – Degrees in Biology and Biochemistry, Department of Life Sciences, University of Coimbra (3x2h). Coord.: João Loureiro.
- ♦ **From 2012 to 2018** Invited lecturer in the advanced course on “Conservation and Management of Biodiversity” of the European Master in Applied Ecology, Department of Life Sciences, University of Coimbra (7x12h). Coord.: Jaime Ramos.
- ♦ **2004** Assistant lecturer in classes of “Entomology”, Degree in Biology, School of Biological Sciences, University of Bristol, UK (1x20). Coord.: Richard Wall.
- ♦ **2005** Assistant lecturer in classes of “Science and success”, Degree in Biology, School of Biological Sciences, University of Bristol, UK (1x10h). Coord.: Jane Memmott.

Weini Huang

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Scientific career and Education

- 01/2018- **Lecturer in Mathematical Biology**, School of Mathematical Sciences, Queen Mary University of London.
Principal Investigator, Theoretical Biology Lab, College of Life Science, Sun Yat-Sen University, Guangzhou, China.
Research topic: tumour and resistance evolution, predator-prey & host-parasite coevolution, stability of ecological communities.
- 03/2016- **Postdoc**, Evolution and Cancer Laboratory, Centre for Tumour Biology, Barts Cancer
12/2017 Institute, Queen Mary University of London.
Research topic: theoretical modelling of tumour dynamics, evolution of drug resistance in ovarian cancer.
- 11/2012- **Postdoc**, Department of Evolutionary Theory, Max Planck Institute for Evolutionary
01/2016 Biology, Plön, Germany.
Research topic: stochastic models on growing populations, impact of trade-offs on population dynamics under coevolution.
- 10/2008- **Dr. rer. nat.**, Research Group for Evolutionary Theory, Max Planck Institute for
08/2012 Evolutionary Biology, Plön, Germany.
Thesis: Evolutionary game dynamics driven by mutations under frequency dependent selection.
- 09/2005- **M.S.**, Ecology, Beijing Normal University, Beijing.
07/2008
- 09/2002- **B.A.**, English, Beijing Normal University, Beijing (Minor).
03/2005
- 09/2001- **B.S.**, Information System and Technology, Beijing Normal University, Beijing (Major).
07/2005

Publications

- **W. Huang***, A. Traulsen, B. Werner, T. Hiltunen, L. Becks, 2017. Dynamical trade-offs arise from antagonistic coevolution and decrease intraspecific diversity. *Nature Communications*, 8:2059.
- A. Baker, **W. Huang**, et al., 2017. Robust RNA-based in situ mutation detection delineates colorectal cancer subclonal evolution. *Nature Communications*, 8:1998.
- A. Amado, L. Fernández, **W. Huang**, F. F. Ferreira, P. R. A. Campos*, 2016. Competing metabolic strategies in a multilevel selection model. *Royal Society Open Science*, 3:160544.
- **W. Huang***, P. R. A. Campos*, V. M. de Oliveira, F. F. Ferreira*, 2016. A resource-based game theoretical approach for the paradox of the plankton. *PeerJ*, 4:e2329.
- **W. Huang**, C. Hauert, A. Traulsen*, 2015. Stochastic game dynamics under demographic fluctuations. *Proceedings of the National Academy of Sciences*, 112:9064-9069.
- A. Amado, **W. Huang**, P. R. A. Campos, F. F. Ferreira*, 2015. Learning process in public goods games. *Physica A: Statistical Mechanics and its Applications*, 430:21-31.
- **W. Huang**, B. Werner, A. Traulsen*, 2012. The impact of random frequency-dependent mutations on the average fitness. *BMC Evolutionary Biology*, 12:160.
- **W. Huang**, B. Haubold, C. Hauert, A. Traulsen*, 2012. Emergence of stable polymorphisms driven by evolutionary games between mutants. *Nature Communications*, 3:919.
- X. Luan*, G. Sun, Y. Qu, **W. Huang**, D. Li, S. Liu, B. Wu, 2012. Prioritizing biodiversity in conservation planning based on C-Plan: a case study from northeast China. *Acta Ecologica Sinica*, 32(3):715~722 (In Chinese with English abstract available).
- **W. Huang**, A. Traulsen*, 2010. Fixation probabilities of random mutants under frequency dependent selection. *Journal of Theoretical Biology*, 263(2):262~268.
- X. Luan*, **W. Huang**, D. Li, M. Liu, 2009. Identification of hotspots and gaps for biodiversity conservation in the Northeast of China based on a systematic conservation planning methodology. *Acta Ecologica Sinica*, 29(1):144~150 (In Chinese with English abstract available).
- L. Zhang*, F. Liu, **W. Huang**, J. Song, 2008. Occurrence of Nematode parasites in raptors in Beijing, China. *Journal of Raptor Research*, 42(3):204~209.

* corresponding authors

Reviewer Activities

Journal of Theoretical Biology, Physical Review Letters, Physical Review E, PNAS, Proc. Royal Soc. B, Proc. Royal Soc. Interface, PLoS Computational Biology, Nature Communications, American Naturalist, BMC Evolutionary Biology, Annals Of Oncology, Gene, Evolution Letters, Evolutionary Applications
FWO, Research Foundation - Flanders in Belgium

Awards and scholarships

- 2016 The 6th Integrated Mathematical Oncology Workshop "Resistance", Research team award, Moffit Cancer Center, Tampa, US
- 2016 1000 talented youth program, Ministry of Education, China
- 2014 Best presentation award, ECMTB 2014, European Conference on Mathematical and Theoretical Biology, Gothenburg, Sweden.
- 2012-2015 Postdoc fellowship, Max Planck Institute, Germany
- 2008-2012 PhD scholarship, China Scholarship Council
- 2008-2009 International fellowship, American Association of University Women Educational Foundation

Organized workshops

- 07/2017 Foresting cancer evolution symposium, Mathematical Models in Ecology and Evolution, London, Germany.

- 10/2010 First workshop Gen-AG 'evolutionary genetics': from genes to ecosystems, German Society of Genetics, Kappelsberg, Germany.

Presentations

Invited

- 05/2018, 15th Annual Internal Symposium at the Max Planck Institute for Evolution Biology - Aquavit 2018, Plön, Germany (Guest scientist in the best presentation award committee; Talk: Evolutionary dynamics with tradeoffs).
- 04/2018, Xiamen University, School of Mathematics, Xiamen, China (Talk: Spatial patterns in cancer evolution).
- 12/2017, East China Normal University, School of Environmental Science, Shanghai, China (Talk: From species coevolution to cancer evolution).
- 10/2017, DynaTrait International Conference, Hannover, Germany (Talk: Dynamical trade-offs under coevolution).
- 03/2015, Graduate school of Gulbenkian Institute of Science, Lisbon, Portugal (Mini-course: Evolutionary Game Theory as one part of the Evolution course, 1 lecture, 3h/lecture)
- 10/2014, Center for Models of Life, Niels Bohr Institute, Copenhagen, Denmark (Talk: Stochastic game dynamics in populations with changing size).
- 02/2014, Summer school of Department of Physics, Federal University of Pernambuco, Recife, Brazil (Mini-course: Evolutionary Game Theory, 3 lectures, 1.5h/lecture)
- 05/2011, International symposium for biodiversity and theoretical ecology, Guangzhou, China (invited talk: Polymorphisms emerges from evolutionary mutant games).

Others

- 12/2017, The International Society for Evolution, Ecology and Cancer, Tempe, US (Talk: Exploring the evolution of cancer and therapy resistance by theory and experiments).
- 07/2017, Mathematical Models in Ecology and Evolution, London, UK (Talk: Revealing mutant fitness and timing by spatial mixing of sub-clones in tumour).
- 07/2016, Evolutionary Biology and Ecology of Cancer Summer School, Cambridge, UK (poster).
- 05/2015, Modelling Biological Evolution: Linking Mathematical Theories with Empirical Realities, Leicester, United Kingdom (Talk: Modelling frequency dependent interactions between types under demographic fluctuations).
- 06/2014, European conference on mathematical and theoretical biology, Gothenburg, Sweden (Two talks: 1. Evolutionary game dynamics under demographic fluctuations; 2. A resource-based model of multi-species competitions under mutations).
- 08/2013, Models in Population Dynamics and Ecology, Osnabrück, Germany (Talk: Stable polymorphisms driven by evolutionary games between mutants).
- 03/2013, 77th annual meeting of the DPG (German physics society) and DPG spring meeting, Regensburg, Germany (Talk: Evolutionary game dynamics under random mutations).
- 07/2011, Summer school on dynamical models in life sciences organized by European society for mathematical and theoretical biology, Evora, Portugal (poster).
- 09/2009, Evolution of cooperation: models and theories, Laxenburg, Austria (poster).

Field work experiences

- 08/2006-10/2006 Population survey of elephants in the tropical forests of Xishuanbanna, Yunnan Province, China
- 07/2006-08/2006 Bird survey in the north-west mountain area in Sichuan Province, China

Other experiences

- 10/2007 Conservation International, Investigation of illegal trade of Chinese Pangolin (*Manis pentadactyla*) in Jiangxi province, China
- 04/2005-06/2006 Roots & Shoots (The Jane Goodall China), Volunteer & group leader of animal welfare and room enrichment team, Beijing Zoo

09/2005- Beijing Raptor Rescue Center, Volunteer & keeper
02/2006

2004 WWF/ Friend of Nature, Volunteer & environment investigator